

Multiscale Hypergraph Masked Autoencoder with Δ -Property Alignment for Novel Molecular Representation Learning

Ziyan Zhu, Yijie Wang, Xian Wei,* and Yanmin Zhang*



Cite This: *J. Chem. Inf. Model.* 2026, 66, 3858–3877



Read Online

ACCESS |



Metrics & More

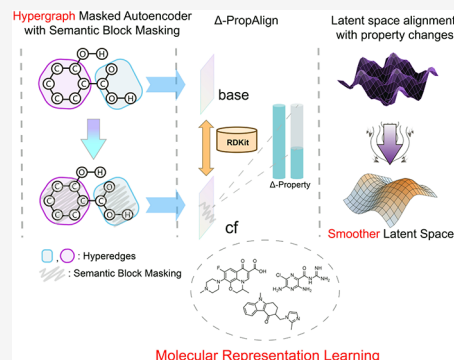


Article Recommendations



Supporting Information

ABSTRACT: Molecular representation learning often focuses on recognizing local structural patterns but struggles to capture functional groups and cross-scale interactions in a chemically meaningful way. Here, we propose the Multiscale Hypergraph Convolutional Masked Autoencoder (MSHG-MAE), a hypergraph-based self-supervised framework for drug-like molecular representation learning. We model each molecule as a unified molecular hypergraph with atom nodes and multitype hyperedges, including bond, ring, functional group, conjugated system, and hydrogen bond hyperedges, and use multiscale hypergraph convolutions to jointly capture dependencies at the atomic, substructural, and molecular levels. In pretraining, a semantics-aware masked autoencoding objective applies biased masking to atoms and hyperedges and reconstructs masked node features and hyperedge attributes, encouraging the model to internalize functional-group semantics. Furthermore, we introduce Δ -Property Alignment (Δ -PropAlign), which aligns embedding differences with proxy property differences so that the learned representations remain sensitive to interpretable structure–property changes. Experiments on multiple public molecular property benchmarks under scaffold splits show that MSHG-MAE consistently outperforms baselines on regression tasks and achieves competitive results on classification benchmarks. On three physicochemical regression benchmarks under scaffold splits (ESOL, FreeSolv, and Lipophilicity), MSHG-MAE with Δ -PropAlign achieves RMSEs of 0.465, 0.780, and 0.501, respectively, corresponding to approximately 40% and 47% lower RMSE than Uni-Mol on ESOL and FreeSolv and approximately 27% lower RMSE than D-MPNN on Lipophilicity (Table 5, scaffold split 80/10/10, 3 seeds), while not degrading structural or functional-group geometry. These results indicate that Δ -PropAlign improves the consistency between embedding differences and property differences without sacrificing structural or functional-group organization in the learned representations. The core code is publicly available on GitHub (<https://github.com/Irzos/MSHG-MAE>), and the data sets used in this study are publicly available from ZINC20 and MoleculeNet.



1. INTRODUCTION

The discovery and development of new drugs is a lengthy and capital-intensive endeavor, typically requiring more than a decade from initial discovery to market approval and relying heavily on extensive experimentation and iterative optimization.^{1,2} To mitigate trial-and-error costs in the early stages of this pipeline, computational models for molecular property prediction have become a key component of modern drug discovery workflows.^{3–6} However, the ultimate performance ceiling of such predictive models is largely determined by the quality and generalizability of the underlying molecular representations.^{3–6} In industrial drug discovery, traditional fixed molecular representations such as extended-connectivity fingerprints⁷ (ECFPs) are widely adopted because of their computational efficiency and robustness. However, these hand-crafted fingerprints have intrinsic limitations in capturing cross-scale interactions spanning atoms, substructures, and whole molecules and in finely characterizing chemically meaningful units such as functional groups.⁶ In recent years, deep molecular representation learning based on molecular graphs, particularly graph neural networks (GNNs), has achieved

performance that rivals or surpasses traditional fingerprints on multiple benchmark data sets, partially alleviating the challenge of integrating local and global structural information.⁸ For example, the supervised AttentiveFP model leverages experimental data together with graph attention mechanisms to improve both predictive performance and interpretability,⁹ whereas directed message passing neural networks (D-MPNNs) and the associated Chemprop software suite have demonstrated strong transferability across public benchmarks and proprietary industrial data sets.^{4,10}

Despite the substantial progress of supervised learning in molecular property prediction, these models remain heavily dependent on expensive and scarce experimentally annotated

Received: December 5, 2025

Revised: March 3, 2026

Accepted: March 5, 2026

Published: March 16, 2026



Table 1. What We Add vs What We Do Not Add Summary^a

Dimension	What we ADD (this work)	What we DO NOT ADD (prior/standard)
Representation	Semantic molecular hypergraph (bond, ring, FG, conj, HBond)	Not claiming novelty in molecular hypergraph/high-order relation modeling per se (recent molecular hypergraph/high-order methods ^{21–26}), we build upon existing hypergraph message-passing frameworks.
SSL paradigm	MAE on hypergraph + joint node + hyperedge reconstruction	Not a generic MAE paradigm (e.g., GraphMAE/2 ^{15,16}), we specialize MAE to molecular hypergraphs with joint node and semantic hyperedge attribute reconstruction.
Corruption	Semantic block masking + leakage control (mask incident hyperedges)	Not a contrastive objective (e.g., MolCLR ¹¹). We use reconstruction with semantic block masking and leakage control.
Property linkage	Δ -PropAlign: align Δ -Embedding with proxy Δ -Property from standard RDKit additive descriptors over masked region	Not task-specific pretext classification (e.g., GROVER ²⁰). We use a single weakly supervised Δ -alignment based on standard RDKit descriptors.

^aNote: Hyperedge types: bond (covalent bond), ring (ring membership), FG (functional group), conj (conjugated substructure), HBond (hydrogen bond).

data.¹¹ To alleviate this bottleneck, self-supervised learning (SSL)-based pretraining has emerged as a dominant paradigm for molecular representation learning.¹² Existing SSL approaches can be broadly categorized into several families. Contrastive methods such as MolCLR leverage graph augmentations to boost downstream performance,¹¹ whereas GraphMVP enforces 2D–3D multiview consistency to enhance geometric sensitivity.¹³ Masked autoencoding (MAE)-style methods, first popularized in vision by He et al. (2022),¹⁴ have since been adapted to graph learning. Representative examples such as GraphMAE and GraphMAE2^{15,16} emphasize robust feature reconstruction rather than explicit structure reconstruction. Chemical language models such as ChemBERTa and Mole-BERT perform masked language modeling pretraining on SMILES strings or context-aware atom vocabularies.^{17–19} Finally, knowledge-enhanced graph transformers such as GROVER incorporate chemically motivated semantic pretext tasks into pretraining to inject domain knowledge.²⁰ However, most of the above approaches are still built on standard pairwise graph structures, which makes it difficult to explicitly represent the ubiquitous multibody interactions and higher-order relations in molecules,^{21–23} such as ring systems, conjugated networks, and hydrogen bond networks.^{24,25} To overcome this limitation, hypergraph-based modeling has attracted growing attention by allowing each hyperedge to connect an arbitrary number of nodes; molecular hypergraphs provide a natural way to encode chemically meaningful “semantic units” such as functional groups and ring systems.^{24–26} Recent work, exemplified by Molecular Hypergraph Neural Networks, constructs high-order message-passing architectures in which hyperedges correspond to functional groups or conjugated structures and reports promising performance on large-scale quantum chemistry benchmarks such as PCQM4Mv2.²⁴ These advances suggest that introducing high-order structures and chemically informed semantic awareness into pretraining objectives is an effective route to further improving molecular representations.

Despite this progress, current molecular representation learning still faces three core bottlenecks. First is a representation bottleneck. Most models operate on standard pairwise molecular graphs, which cannot explicitly encode higher-order many-body relations such as ring systems and conjugated substructures, and existing hypergraph-based architectures only partially exploit their higher-order modeling capacity during pretraining.^{24,25} Second is a semantic bottleneck. Mainstream pretraining objectives, such as uniformly random node, edge, or token masking and generic graph augmentations, are largely decoupled from chemical semantics.²⁷ Masking operations often break apart basic chemical

units such as functional groups and ring systems,²⁸ so the encoder is optimized to reconstruct or discriminate chemically implausible fragments, leading to fragmented and less interpretable representations.¹⁹ Finally, there is an objective bottleneck. Common self-supervised targets (e.g., structural reconstruction, context prediction, or instance discrimination) rarely impose an explicit alignment between structure and properties.²⁹ The model is not required to learn how local structural perturbations induce global property changes, which in turn weakens both the interpretability of the learned representations and their robustness when transferred to property-centric downstream tasks.^{30,31}

To address these challenges, we propose MSHG-MAE (Multiscale Hypergraph Convolutional Masked Autoencoder), a new hypergraph-based masked autoencoder framework. The core idea is threefold. First, we construct a multiscale molecular hypergraph that explicitly models functional groups, ring systems, conjugated systems, and hydrogen bond interactions as semantic hyperedges and employ a multiscale hypergraph convolutional network (MSHG-Conv) to jointly capture complex dependencies at the atomic, substructural, and molecular levels. Second, during pretraining, we adopt a semantics-aware block masking (Block Masking) strategy that masks and reconstructs contiguous blocks of atoms, bonds, and semantic hyperedges. This forces the model to recover the complete semantics of functional groups rather than isolated fragments and prevents it from trivially inferring masked units through shortcut high-order relations in the hypergraph. To bridge the gap between latent feature learning and explicit structure–property alignment, we introduce a complementary pretraining objective Δ -PropAlign. Drawing inspiration from perturbation analysis in medicinal chemistry—where chemists modify functional groups to probe structure–activity relationships (SAR)—this module compels the encoder to model how local structural edits drive global property shifts. Specifically, we treat the masking of semantic blocks not as random noise but as a simulation of a local structural perturbation. We perform two forward passes, including one on the original molecule (the “ground state”) and one on the masked variant (the counterfactual state). The difference between their graph-level embeddings, denoted as Δ -Embedding, represents the model’s sensitivity to the induced structural perturbation in the masked region. Simultaneously, we compute additive atom- or fragment-wise descriptors via RDKit³² and aggregate them over the masked region to derive a proxy property difference (Δ -Property). The Δ -PropAlign objective then constrains the Δ -Embedding to align with the Δ -Property in both direction and magnitude, effectively grounding the learned representation shifts in interpretable, chemically meaningful property

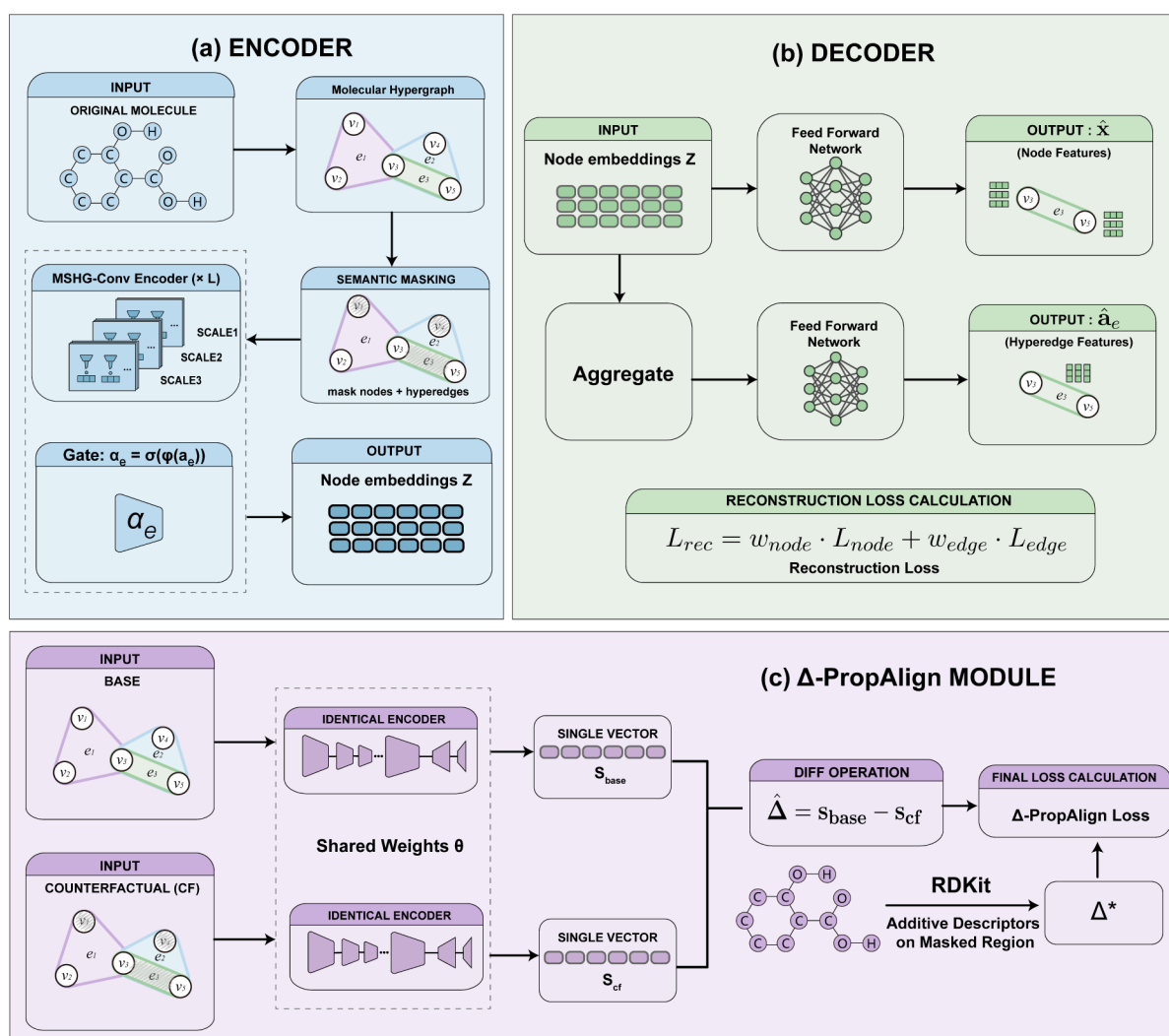


Figure 1. Overview of the proposed MSHG-MAE pretraining architecture and the Δ -PropAlign continued training module. The top row illustrates the masked autoencoding pipeline, including the encoder in (a) and the decoder in (b). The bottom row illustrates the Δ -PropAlign module in (c). (a) Starting from an input molecule, we construct a semantic molecular hypergraph and apply semantic block masking to selected nodes together with their incident hyperedges. A multiscale MSHG-Conv encoder uses an attribute-gated attention coefficient α_e to modulate message aggregation and produces node embeddings Z . (b) Given Z , the decoder reconstructs masked node features $\hat{x}$ and masked semantic hyperedge attributes $\hat{a}_e$ using two feed-forward heads. The reconstruction objective is L_{rec} . (c) For Δ -PropAlign, the encoder with shared weights θ encodes the base molecule and its masked counterfactual, yielding graph-level descriptor vectors s_{base} and s_{cf} . Their difference $\hat{\Delta} = s_{base} - s_{cf}$ is aligned to a proxy property shift Δ^* computed from additive RDKit descriptors aggregated over the masked region.

changes. Although the supervision signals are derived from additive descriptors, this objective fundamentally serves as a sensitivity training mechanism. It regularizes the latent manifold, ensuring that structural perturbations in chemically active regions (functional groups) induce proportional and directionally consistent shifts in the embedding space—a capability that transfers well to complex, nonadditive downstream tasks.

In summary, MSHG-MAE optimizes molecular representations through two complementary pretraining objectives. The semantic masking MAE objective encourages the encoder to reconstruct semantic blocks and higher-order relations at the structural level, effectively learning how to recover chemically meaningful substructures and their interactions. In contrast, the Δ -PropAlign objective operates at the property level, driving the model to capture both the direction and magnitude of property changes induced by local structural edits. Together, these objectives are designed to yield molecular representa-

tions that are more chemically meaningful, more interpretable, and more robustly transferable across property prediction tasks. To clarify the boundary between our contributions and prior work (including GraphMAE/GraphMAE2,^{15,16} GROVER,²⁰ MolCLR,¹¹ and recent molecular hypergraph/high-order relation methods^{21–26}), Table 1 summarizes what we add and what we do not add.

2. METHODS

An overview of the proposed MSHG-MAE and Δ -PropAlign frameworks is shown in Figure 1.

2.1. Notation and Hyperedge Construction

In this work, molecules are first standardized and featurized from SMILES³³ strings and then represented as a unified molecular hypergraph $\mathcal{H} = (\mathcal{V}, \{\mathcal{E}^{(r)}\}_{r \in \mathcal{R}})$. Here, $\mathcal{V} = \{1, \dots, N\}$ denotes the set of atom nodes, and $\mathbf{x}_i \in \mathbb{R}^{d_x}$ is the feature vector of atom i . For any relation type r , a hyperedge $e \in \mathcal{E}^{(r)}$ is associated with an initial

attribute vector $\mathbf{a}_e^{(r)} \in \mathbb{R}^{d_e^{(r)}}$, which is mapped by a relation-specific linear projection $\mathbf{P}^{(r)}$ into a shared d_e -dimensional space, $\tilde{\mathbf{a}}_e = \mathbf{P}^{(r)} \mathbf{a}_e^{(r)} \in \mathbb{R}^{d_e}$. Node features cover basic atomic properties such as element type, charge, and hybridization state (see [Supporting Information, Section S4.1](#) for node features and [Section S4.2](#) for hyperedge attributes). The overall hyperedge set $\mathcal{E} = \{e_1, \dots, e_E\}$ is organized around chemically meaningful “semantic relations”. We treat bond hyperedges as binary relations corresponding to covalent bonds, and their attributes encode bond order, aromaticity/conjugation flags, and related properties. Ring hyperedges are defined using RDKit Chem.GetSymmSSSR (symmetrized SSSR), each returned ring defines one ring hyperedge. This construction is consistent with classical ring-perception and cycle-basis formulations,^{34,35} using RDKit’s symmetry-adapted ring decomposition and connect atoms in the same ring. Their attributes include ring size, aromaticity, and simple topological statistics such as heterocycle indicators and aggregate atomic/charge/hydrogen counts. Functional Group (FG) hyperedges are constructed from template-based functional group definitions specified as curated SMARTS patterns ([Table 2](#), with annotations in [Supporting Information, Section S3](#),

Table 2. SMARTS Library Used to Instantiate FG Hyperedges

carboxyl	[CX3](=O)[OX2H1]	thiol	[SX2H]
hydroxyl	[OX2H]	ketone	[#6][CX3](=O)[#6]
amino	[NX3;H2,H1;!\$(NC=O)]	aldehyde	[CX3H1](=O)[#6]
nitro	[NX3](=O)=O	halogen	[F,Cl,Br,I]
cyano	[CX2]#N	benzene	c1ccccc1
sulfonyl	[SX4](=[OX1])(=[OX1])([OX2H,OX1H0])	pyridine	c1ccncc1
phosphate	[PX4](=[OX1])([OX2H0])([OX2H0])([OX2H0])	imidazole	c1c[nH]cn1
amide	[NX3][CX3](=[OX1])	pyrimidine	c1cncnc1
ether	[OD2](=[#6])[#6]	purine	c1nc2[nH]cnc2n1
ester	[#6][CX3](=O)[OX2][#6]		

[Table S1](#)) and matched using RDKit for common moieties (e.g., carboxyl, hydroxyl, amino, nitro, heteroaromatics). Their attributes include an FG-type embedding together with coarse structural statistics (atom count, aromaticity, average degree, formal charge, and hydrogen count), which implicitly reflect net charge distribution and hydrogen bond capacity. The full list of node features and hyperedge attributes is provided in the [Supporting Information](#). For hydrogen bond hyperedges (HBond), 3D conformations are first optimized with the MMFF³⁶ force field. We then use RDKit ChemicalFeatures to identify donor and acceptor atoms. For each noncovalently bonded donor–acceptor pair, the hydrogen bonded to the donor is located, and a triplet $[D, H, A]$ is assembled. On the resulting 3D conformer, $[D, H, A]$ is classified as an HBond if it simultaneously satisfies three geometric criteria: $H \cdots A$ distance $\leq d_{HA}^{\text{thr}}$, $D \cdots A$ distance $\leq d_{DA}^{\text{thr}}$, and $\angle D-H \cdots A \geq \theta^{\text{thr}}$. Unless otherwise specified, we adopted a relaxed hydrogen bond criterion inspired by the classical geometric definitions of Baker & Hubbard (1984) and McDonald & Thornton (1994), allowing slightly larger distance thresholds, $d_{HA}^{\text{thr}} = 3.5 \text{ \AA}$, $d_{DA}^{\text{thr}} = 4.0 \text{ \AA}$, and $\theta^{\text{thr}} = 120^\circ$.^{37,38}

In all experiments, a single ETKDG + MMFF conformer is generated per molecule as an offline preprocessing step, and the resulting HBond hyperedges are used consistently in both pretraining and all downstream training and evaluation. Conjugated (Conj) hyperedges capture conjugated systems, which we identify as connected components of RDKit-annotated conjugated bonds; their attributes encode global quantities such as conjugation length and aggregate electronic/structural statistics, placing different conjugated

fragments into a common representation space. In summary, we instantiate hyperedges for relations originating from different sources and scales and embed all hyperedge attributes into the same d_e -dimensional space (e.g., ring size and topology, functional-group type embeddings, geometric statistics of hydrogen bonds, and aggregate structural/electronic descriptors). Implementation details of hyperedge construction and feature templates are provided in the [Supporting Information \(Sections S1, S3, and S4\)](#). Data set-level hyperedge statistics are provided in the [Supporting Information \(Section S10 and Tables S4–S6 and Figures S3–S5\)](#). For clarity and reproducibility, we summarize the deterministic instantiation rules below and list the SMARTS library used for FG hyperedges in [Table 2](#). This unified design allows the model to simultaneously observe multigranular dependencies from atoms to fragments to whole molecules within a single graph domain and to perform message passing and attention through a consistent tensor interface, thereby avoiding the engineering overhead and information fragmentation associated with multigraph coupling or cascaded architecture.

Rings are enumerated by RDKit Chem.GetSymmSSSR (symmetrized SSSR); each returned ring defines one ring hyperedge. Fused systems yield multiple overlapping rings; duplicates are removed by atom-set identity, and rings with <3 atoms are discarded. Conjugated bonds follow RDKit’s conjugation perception; we build the subgraph induced by conjugated bonds and take each connected component as one Conj hyperedge; components with <3 atoms are discarded. FG hyperedges are detected by matching a curated SMARTS library. For each SMARTS hit, we apply a fixed one-hop neighborhood expansion (depth = 1), deduplicate by atom-set identity, and discard expanded hits with <2 atoms. HBond donors/acceptors are identified by RDKit ChemicalFeatures (BaseFeatures.fdef). Covalently bonded donor–acceptor pairs are excluded. A triplet $[D, H, A]$ is retained as an HBond hyperedge only if it satisfies the geometric criteria $H \cdots A \leq d_{HA}^{\text{thr}}$, $D \cdots A \leq d_{DA}^{\text{thr}}$, $\angle D-H \cdots A \geq \theta^{\text{thr}}$, with $d_{HA}^{\text{thr}} = 3.5 \text{ \AA}$, $d_{DA}^{\text{thr}} = 4.0 \text{ \AA}$, and $\theta^{\text{thr}} = 120^\circ$.

Simple examples of hyperedge construction: In acetic acid (CC(=O)O), the SMARTS pattern [CX3](=O)[OX2H1] matches a carboxyl functional group. In naphthalene, Chem.GetSymmSSSR returns the two smallest rings, producing two overlapping ring hyperedges. In 1,3-butadiene (C=CC=C), conjugated bonds form a single connected component (via RDKit bond.GetIsConjugated), thus, one conjugated-system hyperedge. For salicylic acid (O=C(O)c1ccccc1O), on the RDKit-embedded conformer (randomSeed = 42), we detect one intramolecular hydrogen bond hyperedge $[D, H, A]$ with $H \cdots A \approx 1.70 \text{ \AA}$, $D \cdots A \approx 2.56 \text{ \AA}$, and $\angle D-H \cdots A \approx 144.5^\circ$.

2.2. Semantic Block Masking

For pretraining data construction, we adopt a semantics-aware block masking strategy, termed Semantic Block Masking. Concretely, for each molecule, we first decompose its hypergraph into chemically meaningful fragments, including functional groups, ring systems, and short acyclic chain segments, identified by RDKit pattern matching and ring/chain analysis; this pool of fragments defines the candidate semantic blocks. At each training step, the model samples a subset of these fragments and takes the union of their atoms as the semantic part of the masked region and further adds uniformly random atoms so that approximately 70% of the masked atoms (by count) come from semantic blocks, while the remaining $\approx 30\%$ are sampled at random, which increases the difficulty of the reconstruction task and encourages the network to learn contextual representations that are coherent with both local chemical semantics and their surrounding environment. To prevent the model from trivially recovering masked atoms solely from higher-order relations, once an atom is masked, all hyperedges incident on that atom (including bond and semantic hyperedges) are simultaneously set to the masked state in that step. Compared with purely random node masking, this semantics-aware strategy is more faithful to chemical reality and consistently increases both the information content and the difficulty of the reconstruction objective. During the initial pretraining stage, the masking ratio is fixed at 0.70, and in Δ -PropAlign, the masking ratio is annealed from

0.40 to 0.25 using a cosine decay schedule. These choices are supported by systematic ablations over semantic to random mixing ratios, MAE mask ratios, and Δ -PropAlign schedules reported in the Supporting Information (Section S11, Tables S8–S10 and Figures S6–S8).

2.3. Encoder

The encoder adopts a multi-receptive-field hypergraph convolution that constructs several diffusion scales on the same underlying hypergraph. Let $\mathbf{H} \in \{0,1\}^{|V| \times |E|}$ denote the node–hyperedge incidence matrix. For the first scale, we use the original incidence, $\mathbf{H}^{(1)} = \mathbf{H}$. For higher scales $s > 1$, we obtain effective incidence matrices by applying $s - 1$ sparse diffusion steps of the form $\mathbf{H} \leftarrow \mathbf{H}(\mathbf{H}^T \mathbf{H})$ on the vertex–hyperedge bipartite structure and then binarizing the resulting matrix to get $\mathbf{H}^{(s)}$. Intuitively, $\mathbf{H}^{(s)}$ connects each node to hyperedges that become reachable after $s - 1$ rounds of hypergraph diffusion (alternating node–node and node–hyperedge interactions), thereby capturing higher-order comembership patterns beyond the original hyperedges. In practice, we cap the diffusion depth to a small maximum (e.g., three steps) to preserve sparsity and numerical stability.

For each scale s , we derive a sparse hyperedge index from $\mathbf{H}^{(s)}$ and apply the same vectorized hypergraph convolution operator to update node features. Concretely, scale s first applies a learned linear projection $\mathbf{W}_s$ to the input node features, followed by one round of message passing in which transformed node features are aggregated to hyperedges and scattered back to nodes according to $\mathbf{H}^{(s)}$. The outputs from all scales are then stacked and fused by lightweight mean aggregation across scales and passed through a shared output projection together with residual and normalization blocks, yielding stable deep representations with scale-dependent receptive fields on the same hypergraph. A full matrix-form description of the single-scale operator and normalization choice is provided in Supporting Information (Section S2).

To encode coarse structural information, we concatenate a lightweight positional encoding (normalized node degree and an approximate centrality score) to the node features once at the first layer, rather than injecting additional positional or graph-level signals into every layer, which empirically reduces the distribution shift. Global molecular-level embeddings are obtained by mean pooling the final node representations and are used only in auxiliary graph-level prediction heads. Beyond standard message passing, we further use hyperedge attributes as gating signals to implement attention-enhanced updates of the node representations. Specifically, the hyperedge attribute gate produces a scalar coefficient α_e as defined in the following equation (eq 1):

$$\alpha_e = \sigma(\phi_w(\mathbf{a}_e)) \in (0, 1) \quad (1)$$

where ϕ_w denotes a small MLP, σ is the sigmoid function, and $\alpha_e \in (0,1)$ is the weight produced by the attribute gate. We first map the hyperedge attributes to a scalar weight in $(0,1)$ through this small MLP and then aggregate the self-attention-refined hyperedge vectors into their incident nodes according to the corresponding gating weights.

$$\mathbf{m}_i = \sum_{e \in N(i)} \alpha_e \mathbf{g}_e \quad (2)$$

In eq 2, $\mathbf{g}_e$ denotes an attribute-refined representation vector of the hyperedge and $\mathbf{m}_i$ is the aggregated message vector of the node. This design implements an attribute-driven, structure-aware weighted message aggregation. It introduces only a light computational overhead that scales linearly with the number of hyperedges, yet markedly improves the model's sensitivity to key semantic relations. On the decoder side, node and hyperedge attributes are reconstructed in parallel. The node head regresses atom-level features from the hidden representations of masked nodes, whereas the hyperedge head aggregates first the incident nodes to obtain the hyperedge representation and then regresses its attribute vector. The overall pretraining objective is dominated by a combination of node

reconstruction and hyperedge reconstruction, with L_{node} and L_{edge} defined in the following equation (eq 3):

$$L_{\text{node}} = \frac{1}{|\mathcal{V}_{\text{mask}}|} \sum_{i \in \mathcal{V}_{\text{mask}}} l(\hat{\mathbf{x}}_i, \mathbf{x}_i), \quad L_{\text{edge}} = \frac{1}{|\mathcal{E}_{\text{mask}}|} \sum_{e \in \mathcal{E}_{\text{mask}}} l(\hat{\mathbf{a}}_e, \mathbf{a}_e) \quad (3)$$

where l is instantiated as mean squared error (MSE) for continuous attributes. The node-reconstruction loss is evaluated only on masked nodes, and the hyperedge-reconstruction loss is evaluated only on semantic hyperedges that are synchronously masked for “leakage protection”, so that the training signals are strictly aligned with the masking decisions. In practice, we simply use the same MSE loss for both node- and hyperedge-level reconstruction.

2.4. Loss Function and Training Objective

During pretraining, the overall objective comprises a node reconstruction loss and a hyperedge reconstruction loss. Due to the heterogeneous nature of atomic features and hyperedge attributes, their respective loss terms exhibit inherent scale discrepancies, leading to gradient magnitude mismatch. Consequently, employing static weights often results in optimization instability or causes one task to prematurely dominate the learning process. Drawing inspiration from established adaptive weighting paradigms, specifically the gradient equalization of GradNorm and the loss-trend adaptation of Dynamic Weight Average (DWA),^{39,40} we design a streamlined balancing strategy named the Target Contribution Controller (TCC). This mechanism dynamically monitors the relative training progress to adjust task weights, thereby obviating the need for computationally expensive manual tuning or large-scale hyperparameter sweeps to determine the optimal static values. The reconstruction objective is defined in the following equation (eq 4):

$$L_{\text{rec}} = w_{\text{node}} L_{\text{node}} + w_{\text{edge}} L_{\text{edge}} \quad (4)$$

The weights are adjusted online by TCC. Periodically, we compute the contribution ratio of each component as follows (as in eq 5):

$$\rho_{\text{node}} = \frac{w_{\text{node}} \overline{L_{\text{node}}}}{w_{\text{node}} \overline{L_{\text{node}}} + w_{\text{edge}} \overline{L_{\text{edge}}}}, \quad \rho_{\text{edge}} = 1 - \rho_{\text{node}} \quad (5)$$

The contribution ratios are then exponentially smoothed with EMA. If the smoothed ρ_e of component e drifts outside its prescribed target interval, reflecting the relative complexity of each task, the corresponding weight is iteratively adjusted using a conservative step size in the log domain. This mechanism keeps the training process close to a stable equilibrium. Specific configurations for the intervals, tolerance margins, and step sizes are detailed in the Supporting Information (Section S6, Loss and Dynamic Weighting/TCC).

2.5. Δ -PropAlign

In all Δ -PropAlign experiments, we used a fixed set of five RDKit molecular descriptors: Crippen logP (MolLogP), Crippen molar refractivity (MolMR), topological polar surface area (TPSA), Labute's approximate surface area (LabuteASA), and the electrotopological state index (EState). These descriptors are standard structure-derived quantities computed directly from the molecular graph and do not involve any downstream benchmark labels.

Following the initial reconstruction-based pretraining, we introduce a weakly supervised continued training phase: Δ -PropAlign, with the objective given in eq 6:

$$L_{\text{cont}} = L_{\text{rec}} + \lambda_{\Delta} L_{\Delta} + \lambda_{\text{abs}} L_{\text{abs}} \quad (6)$$

where λ_{Δ} follows a linear warm-up schedule up to λ_{max} and λ_{abs} is a small constant. Specifically, we executed two parallel forward passes for the same molecule. The baseline branch takes the unmasked graph to produce the graph-level fingerprint and predicts multidimensional molecular descriptors ($\mathbf{s}_{\text{base}}$). The counterfactual branch applies masking to a set of semantic subgraphs (motifs) to obtain $\mathbf{s}_{\text{cf}}$. The resulting difference, $\hat{\Delta} = \mathbf{s}_{\text{base}} - \mathbf{s}_{\text{cf}}$, represents the model's property sensitivity response to the masked semantic region.

To construct the ground truth for this structural difference, we employ additive atom/fragment descriptors from RDKit to derive a consistency regularization target, Δ^* : the contributions of atoms within the masked subgraph are summed to calculate the expected direction and magnitude of the property change. To eliminate systematic bias arising from variations in the substructure size, we employ a two-level normalization strategy.

First, we normalize both $\hat{\Delta}$ and Δ^* by $\sqrt{k}$ (where k is the motif size). This is grounded in the statistical observation that the cumulative variance of additive properties typically scales linearly with atom count (i.e., standard deviation scales with $\sqrt{k}$). Without this scaling, the loss landscape would be dominated by numerical fluctuations from large scaffolds (e.g., fused ring systems), thereby overshadowing the contributions of small yet chemically reactive functional groups (e.g., hydroxyl or amino groups). By dividing by $\sqrt{k}$, we achieve a “decoupling of semantic importance from structural size,” ensuring that the model learns structure–property responses with equal weight, regardless of whether the feature arises from a massive scaffold or a tiny substituent.

Second, we apply batch-wise Z-score standardization to the target difference Δ^* to unify the dynamic ranges and units of disparate physicochemical properties, eliminating covariate shift and ensuring stability in multitask learning. Finally, a robust Huber loss ($\beta = 1.0$) is applied across valid dimensions for regression. Simultaneously, to stabilize the overall representation scale and bias, a weak absolute anchor loss is applied to the descriptor set $\mathcal{A}$ (MolLogP, TPSA, EState, LabuteASA, MolMR), as defined in eq 7:

$$L_{\text{abs}} = \sum_{d \in \mathcal{A}} \text{MSE}(z(\hat{s}_{\text{base},d}), z(y_d)) \quad (7)$$

where y_d represents the descriptor pseudolabels calculated by RDKit and $z(\cdot)$ denotes standardization. The overall objective of this continued training superimposes the Δ -alignment and absolute anchor terms onto the self-supervised reconstruction loss. Notably, the Δ weight employs an independent linear warm-up schedule, ensuring the model first converges on structural/semantic reconstruction before progressively injecting property alignment signals.

The ablation study conducted across three random seeds (Table 3) demonstrates that the Δ -PropAlign module confers a genuine

Table 3. Performance with 1500 Training Steps (Mean $\pm$ Std, 3 Seeds)

Method	BBBP (AUROC)	ESOL (RMSE)	FreeSolv (RMSE)
Δ -PropAlign	0.9349 $\pm$ 0.02	0.4650 $\pm$ 0.05	0.7800 $\pm$ 0.11
Self-supervised	0.9330 $\pm$ 0.03	0.5350 $\pm$ 0.08	0.8470 $\pm$ 0.16

enhancement to representation quality. Crucially, these performance gains are intrinsic to the proposed alignment strategy and are not merely an artifact of the extended training duration (i.e., an increased number of training steps).

2.6. Training Details and Experimental Configuration

To balance training stability with throughput, we implemented specific engineering configurations at the implementation level. These include the optimizer (AdamW,⁴¹ with weight decay excluded for bias, normalization, and embedding parameters), learning rate schedules (Warmup–Stable–Decay (WSD) scheduler⁴² with independent warmup for the Δ -loss), mixed-precision training (BF16 + FP32), gradient scaling and clipping, and stepwise validation protocols, including early stopping and checkpointing (Supporting Information, Section S6). For the final fingerprint extraction, masking operations are disabled; fixed-length molecular fingerprints are derived solely via graph-level aggregation of the node hidden representations output by the encoder. We employ linear and two-layer MLP prediction heads for classification and regression tasks, respectively, strictly adhering to data splits and evaluation metrics consistent with established baselines. Optional test-time postprocessing (k-nearest-neighbor smoothing, ridge regression, Platt calibration) is applied uniformly

to all models, with full details in the Supporting Information (Section S6). The downstream hyperparameter tuning protocol and the exact search space are reported in Supporting Information (Section S6 and Table S2).

2.7. Complexity

Regarding time and space complexity, the computational cost of the backbone message passing scales linearly with the number of incidence connections and hidden dimensions. The attention-enhanced and multiscale components are implemented using sparse operators and cache reuse, ensuring that GPU memory consumption grows approximately linearly with molecular size. Overall, the framework is engineered to capture multisource semantic relations while maintaining end-to-end trainability. Furthermore, the two-stage “reconstruction-then-alignment” design injects property semantics into the representations without compromising structural understanding, thereby endowing the resulting molecular fingerprints with enhanced robustness in terms of interpretability.

2.8. Data Sets and Splitting Methods

It is well established that scaffold splitting presents a greater challenge than random splitting; in molecular property prediction tasks, scaffold-based splits provide a more rigorous evaluation of a model's out-of-distribution (OOD) generalization ability. We evaluated our model on nine popular benchmarks from MoleculeNet⁴³ using scaffold splits across three random seeds. To ensure a fair comparison, we adopt a unified downstream evaluation protocol for all methods: all models use the same scaffold-based train/validation/test splits (0.8/0.1/0.1), the same random seeds, and the same validation-based checkpoint selection. For 2D graph models implemented in our codebase, we further use the same fine-tuning loop and the same downstream hyperparameter search space; details are provided in Supporting Information (Section S6, Table S2).

2.9. Baseline Models

We benchmarked MSHG-MAE against a diverse set of strong established baselines. All methods are evaluated under the same downstream evaluation protocol, including identical scaffold splits, random seeds, and validation-based checkpoint selection. For 2D graph models implemented and trained in our codebase, we use matched optimization/stopping settings and tune the same set of downstream hyperparameters on the validation split using a shared discrete search space (Supporting Information, Section S6, Table S2). No additional task-specific tuning is applied to our method beyond this shared search space. Unless otherwise stated, downstream results are obtained by end-to-end fine-tuning (no layer freezing); the encoder is frozen only for representation analyses (e.g., t-SNE) and linear-probe experiments. Traditional ML and supervised GNN baselines, including ECFP4 + RF, AttentiveFP, D-MPNN, GCN, and GIN, are trained from random initialization on each downstream data set under this protocol. For GROVER_large, GROVER_base, Uni-Mol, MolCLR, and ImageMol, we use the official pretrained checkpoints released by the original authors and fine-tune them on the downstream tasks with the same evaluation protocol. In particular, Uni-Mol and ImageMol are trained using their native 3D-structure and image-based input pipelines and the recommended optimization settings; we replace only their original train/validation/test splits with our scaffold-based splits. GraphMAE is reimplemented and pretrained from scratch on the same ZINC20-druglike 250k corpus as MSHG-MAE, followed by downstream fine-tuning under the unified protocol. Unless otherwise stated, method-specific hyperparameters that are not shared with MSHG-MAE are kept at their default values in the original implementations.

The selected baselines are categorized as follows:

- Traditional ML: ECFP⁷ + Random Forest (RF)⁴⁴ represents nongraph methods utilizing fixed molecular fingerprints.
- Supervised GNNs: D-MPNN,⁴ GIN,⁴⁵ GCN,⁴⁶ and AttentiveFP⁹ represent classic Graph Neural Network architectures trained in a supervised manner.
- Self-Supervised Pretraining: GROVER-large,²⁰ GROVER-base,²⁰ GraphMAE,¹⁵ MolCLR,¹¹ and Uni-Mol⁴⁷ share a

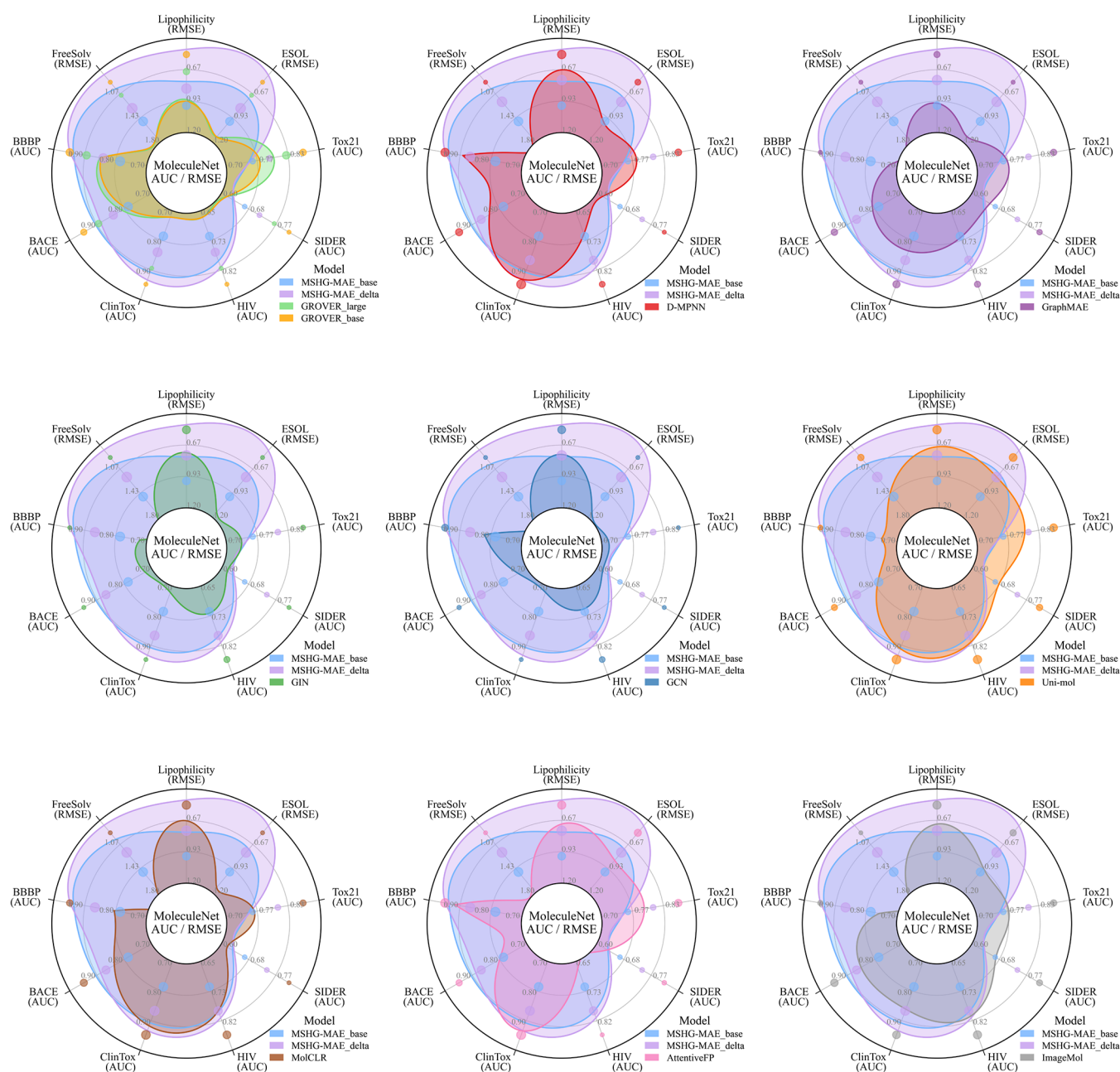


Figure 2. Overall comparison of MSHG-MAE variants and baseline models on nine MoleculeNet benchmarks. For visualization in this figure, the RMSE values are monotonically transformed so that higher scores indicate better performance. All numerical results in Tables 4 and 5 are reported by using the original RMSE values.

conceptual similarity with our approach, falling under the category of self-supervised molecular representation learning.

- **Multimodal/Image-based:** ImageMol⁴⁸ represents a distinct approach that learns molecular representations from visual data (molecular images).

In this work, we benchmark MSHG-MAE against a representative spectrum of baselines spanning fixed fingerprints, supervised graph learning, self-supervised pretraining, and multimodal/image-based representation learning, all under a unified scaffold-split protocol with identical seeds, validation-based checkpoint selection, and matched fine-tuning settings to ensure comparability. Specifically, the traditional ECFP4 + RF baseline is computationally efficient and often robust in small-data regimes, yet its hand-crafted fingerprint representation is intrinsically limited in capturing cross-scale interactions among atoms, substructures, and whole molecules and in finely characterizing chemically meaningful units such as functional

groups. Supervised 2D GNNs (e.g., GCN/GIN/D-MPNN/AttentiveFP) learn task-adaptive representations from molecular topology and can partially alleviate the integration of local and global structural information, but their performance is strongly constrained by the availability of expensive labeled data because these models are trained from random initialization on each downstream data set. Self-supervised pretraining baselines (e.g., MolCLR, GraphMAE, GROVER, and Uni-Mol) improve label efficiency by transferring generic chemical knowledge; however, many existing methods remain built on standard pairwise graphs and generic objectives, which makes it difficult to explicitly represent ubiquitous higher-order relations (rings, conjugation, hydrogen bond networks) and rarely enforces an explicit structure–property alignment, leading to an objective bottleneck when transferring to property-centric tasks. Finally, multimodal or alternative-input methods (e.g., Uni-Mol with 3D conformers and ImageMol with molecular images) can be advan-

Table 4. Performance Comparison on Classification Benchmarks

Model	Classification (higher is better)					
	BBBP	BACE	ClinTox	HIV	SIDER	Tox21
ECFP4 + RF	0.790 ± 0.06	0.844 ± 0.04	0.730 ± 0.01	0.601 ± 0.02	0.540 ± 0.01	0.682 ± 0.13
D-MPNN	0.882 ± 0.04	0.825 ± 0.04	0.927 ± 0.02	0.701 ± 0.10	0.590 ± 0.08	0.766 ± 0.00
GraphMAE	0.703 ± 0.02	0.793 ± 0.01	0.823 ± 0.01	0.712 ± 0.01	0.640 ± 0.01	0.755 ± 0.01
GIN	0.614 ± 0.01	0.702 ± 0.01	0.597 ± 0.13	0.712 ± 0.08	0.571 ± 0.01	0.719 ± 0.01
GCN	0.807 ± 0.00	0.711 ± 0.04	0.695 ± 0.09	0.700 ± 0.03	0.533 ± 0.02	0.700 ± 0.07
GROVER_large	0.840 ± 0.01	0.794 ± 0.02	0.688 ± 0.18	0.522 ± 0.01	0.580 ± 0.02	<u>0.795 ± 0.03</u>
GROVER_base	0.836 ± 0.01	0.778 ± 0.15	0.660 ± 0.01	0.527 ± 0.01	0.586 ± 0.01	0.762 ± 0.02
Uni-Mol	0.731 ± 0.02	0.801 ± 0.00	0.905 ± 0.04	<u>0.800 ± 0.00</u>	0.670 ± 0.01	0.796 ± 0.01
MolCLR	0.790 ± 0.02	0.824 ± 0.01	0.912 ± 0.04	0.781 ± 0.01	0.589 ± 0.01	0.750 ± 0.00
AttentiveFP	0.910 ± 0.01	0.784 ± 0.02	0.930 ± 0.05	0.651 ± 0.09	0.616 ± 0.06	0.784 ± 0.02
ImageMol	0.739 ± 0.00	0.839 ± 0.01	0.851 ± 0.01	0.797 ± 0.00	<u>0.660 ± 0.03</u>	0.759 ± 0.01
MSHG-MAE_base	<u>0.931 ± 0.03</u>	0.897 ± 0.01	0.901 ± 0.09	0.783 ± 0.04	0.618 ± 0.02	0.737 ± 0.03
MSHG-MAE_delta	0.944 ± 0.02	<u>0.878 ± 0.04</u>	<u>0.929 ± 0.02</u>	0.802 ± 0.04	0.623 ± 0.04	0.740 ± 0.01

geous when downstream signals depend strongly on conformation-level geometry or global charge distribution, but they introduce additional preprocessing assumptions and modality-specific inductive biases, which do not necessarily translate into uniform gains across all 2D scaffold-split benchmarks.

Collectively, these baselines span a broad spectrum, ranging from traditional machine learning models to supervised GNNs and self-supervised pretraining methods, and they cover both 2D molecular topologies and 3D geometries. This comprehensive benchmark suite provides a robust and representative reference for evaluating the efficacy of our proposed method.

3. RESULTS

3.1. Main Results

3.1.1. Pretraining Data and Evaluation Setup. We first constructed the pretraining corpus by uniformly selecting 250,000 SMILES strings from the full ZINC20-drug-like database⁴⁹ using reservoir sampling⁵⁰ to ensure equal probability during a single sequential pass (with the random seed fixed at 42). We selected ZINC20 over the historically common ZINC12/15-250 K subsets because it better reflects the currently commercially available chemical space. Notably, we intentionally employed pure random sampling rather than scaffold-aware sampling as the downstream property prediction benchmarks universally utilize scaffold splitting; this separation is crucial to rigorously evaluate generalization. We also performed a rigorous InChIKey⁵¹ zero-intersection check across all data sets to preclude potential data leakage (see Supporting Information, Section S5, for details).

For downstream evaluation, we assessed the model on nine standard benchmark data sets from MoleculeNet using scaffold splits across three random seeds. An overall comparison of MSHG-MAE variants and baseline models across these benchmarks is shown in Figure 2. Experimental results demonstrate that MSHG-MAE matches or surpasses strong baseline models on multiple regression tasks (e.g., ESOL, FreeSolv) while maintaining competitive performance on classification benchmarks. To further validate the specific contribution of the Δ -PropAlign objective, we conducted a comparative analysis on 2,039 molecules from the BBBP data set across 15 random seeds. The results indicate that this module successfully enhances the model's property sensitivity without compromising structural or functional group semantics (evidenced by noninferior unsupervised geometric metrics and stable semantic enrichment results). Specifically, in linear

probe tests with a frozen encoder, we observed marginal increases in mean-AUROC (+0.004) and mean-AUPRC (+0.007), alongside a significant enhancement in semantic distance consistency (significant pairs increased from 25 to 27, Cliff's δ^{52} +0.013). Collectively, these findings suggest that the property sensitivity and robustness injected by Δ -PropAlign align consistently with the observed performance improvements in downstream property prediction tasks.

3.1.2. Metrics and Baselines. Following standard protocols for molecular property prediction and using matched data splits and comparable downstream fine-tuning budgets across methods (details provided in the Supporting Information), we employed ROC-AUC (higher is better) as the evaluation metric for classification tasks (e.g., BBBP, BACE, ClinTox). For regression tasks (e.g., ESOL, Lipophilicity, FreeSolv), we utilized RMSE (lower is better) to assess model performance.

Table 4 summarizes the AUROC performance across six molecular classification benchmarks, including BBBP, BACE, ClinTox, HIV, SIDER, and Tox21. To facilitate interpretation of the comparative results, we categorize the baselines into four groups: (i) fingerprint-based ML (ECFP4 + RF), (ii) supervised 2D GNNs trained from scratch (e.g., GCN/GIN/D-MPNN/AttentiveFP), (iii) self-supervised pretraining on 2D molecular graphs (e.g., MolCLR, GraphMAE, and GROVER), and (iv) 3D/multimodal or image-based representation learning (Uni-Mol and ImageMol). Fingerprint methods are efficient but rely on fixed descriptors with limited task adaptivity; supervised GNNs can learn discriminative patterns but are label-hungry under scaffold splits; 2D self-supervised methods improve label efficiency yet may be constrained by pairwise-graph inductive biases and generic objectives; and 3D/multimodal or image-based methods can capture conformational or global-pattern signals, but their gains are task-dependent and hinge on modality-specific preprocessing. In general, both MSHG-MAE variants consistently rank among the top performers across these baseline categories, suggesting that multiscale hypergraph modeling combined with self-supervised pretraining offers robust adaptability for classification tasks. At the individual data set level, MSHG-MAE_delta achieved AUROC scores of 0.944 on BBBP and 0.802 on HIV, outperforming all baseline models. Notably, on BBBP, MSHG-MAE_delta improved by approximately 0.034 over the best-performing supervised model, AttentiveFP (0.910), and significantly surpassed the traditional

fingerprint method ECFP4 + RF (0.790). Similarly, on HIV, MSHG-MAE_delta slightly outperformed the 3D-informed Uni-Mol (0.800) and also surpassed 2D-pretrained and image-based baselines (e.g., MolCLR/GraphMAE/GROVER and ImageMol), indicating that our hypergraph semantic modeling better captures property-coupled decision cues in this task. These results suggest that introducing Δ -PropAlign enhances discriminative capability particularly relative to fingerprint-based and from-scratch supervised 2D GNN baselines when tasks correlate strongly with physicochemical properties and membrane permeability.

Conversely, on BACE, MSHG-MAE_base achieved the highest score (0.897), showing a substantial advantage over ECFP4 + RF (0.844) and Uni-Mol (0.801), as well as other GNN, 2D-pretrained, and image-based baselines. The performance of MSHG-MAE_delta was comparable to that of the base version, with only a slight decrease. This indicates that the multiscale hypergraph structure itself is sufficient to capture local pharmacophore patterns associated with BACE activity, leaving limited room for additional gains from Δ -PropAlign when the decision boundary is already well encoded by structural motifs. On the ClinTox data set, AttentiveFP and MSHG-MAE_delta yielded nearly identical AUROCs (0.930 vs 0.929), while MSHG-MAE_base scored slightly lower. This confirms that classic attention-based GNNs remain highly competitive in small-sample toxicity tasks, yet MSHG-MAE_delta achieves parity without relying on explicit 3D priors.

Synthesizing these observations, Δ -PropAlign tends to yield additional gains over fingerprint-based and from-scratch supervised 2D GNN baselines on property-coupled classification tasks, while its primary contribution shifts toward robustness when structural information already dominates the decision boundary. For the multilabel toxicity prediction tasks SIDER and Tox21, Uni-Mol retained the best AUROC scores (0.670 and 0.796, respectively), while MSHG-MAE variants trailed behind; notably, ImageMol and some 2D-pretrained baselines are also strong on these benchmarks, suggesting that modality-specific priors and task-specific pretraining signals can be particularly helpful. Unlike the previously discussed tasks, SIDER and Tox21 are characterized by high label imbalance and discrete toxicity modes rather than continuous physicochemical gradients, conferring an advantage to Uni-Mol's 3D conformation and global charge distribution modeling and, in some cases, to image-based representations. Comparing the base and delta versions reveals that incorporating Δ -PropAlign provided negligible improvement or slight regression on these specific data sets, implying that a physicochemical proxy alignment objective may be less aligned with toxicity labels. This suggests that in tasks with extreme class imbalance or multilabel toxicity, the weight of the Δ -alignment loss should be reduced, or a structure-first training strategy adopted, to prevent property proxy objectives from imposing excessive constraints on decision boundaries.

Table 5 presents the RMSE comparison across three physicochemical regression benchmarks, including FreeSolv, ESOL, and Lipophilicity. In contrast to the classification results, MSHG-MAE_delta achieved the top rank on all regression data sets. This underscores that the contribution of Δ -PropAlign is significantly more pronounced in scenarios requiring precise characterization of the continuous physicochemical properties. Specifically, on FreeSolv, MSHG-MAE_delta achieved an RMSE of 0.780, reducing the error

Table 5. Performance Comparison on Regression Benchmarks

Model	Regression (lower is better)		
	FreeSolv	ESOL	Lipo
ECFP4 + RF	4.02 $\pm$ 2.00	1.730 $\pm$ 0.51	1.110 $\pm$ 0.05
D-MPNN	2.080 $\pm$ 0.08	1.030 $\pm$ 0.01	<u>0.688 $\pm$ 0.05</u>
GraphMAE	1.970 $\pm$ 0.08	1.350 $\pm$ 0.02	0.993 $\pm$ 0.12
GIN	2.350 $\pm$ 0.92	1.570 $\pm$ 0.05	0.757 $\pm$ 0.02
GCN	3.170 $\pm$ 0.71	1.210 $\pm$ 0.05	0.773 $\pm$ 0.01
GROVER_large	3.540 $\pm$ 0.40	1.710 $\pm$ 0.16	0.960 $\pm$ 0.04
GROVER_base	3.100 $\pm$ 0.12	1.910 $\pm$ 0.12	0.977 $\pm$ 0.09
Uni-Mol	1.480 $\pm$ 0.09	0.781 $\pm$ 0.01	0.703 $\pm$ 0.11
MolCLR	2.590 $\pm$ 0.25	1.270 $\pm$ 0.04	0.691 $\pm$ 0.00
AttentiveFP	2.070 $\pm$ 0.18	0.877 $\pm$ 0.02	0.721 $\pm$ 0.00
ImageMol	2.020 $\pm$ 0.07	0.970 $\pm$ 0.07	0.721 $\pm$ 0.01
MSHG-MAE_base	<u>1.010 $\pm$ 0.15</u>	<u>0.743 $\pm$ 0.07</u>	0.795 $\pm$ 0.01
MSHG-MAE_delta	0.780 $\pm$ 0.11	0.465 $\pm$ 0.05	0.501 $\pm$ 0.04

by approximately 47% relative to the best baseline, Uni-Mol (1.480). On ESOL, MSHG-MAE_delta lowered the RMSE to 0.465, an $\sim$ 40% reduction compared to Uni-Mol (0.781). On Lipophilicity, the RMSE of 0.501 represented an $\sim$ 27% improvement over D-MPNN (0.688). The consistency of these results across varying data set sizes and property types demonstrates that automatically constructing molecular pairs via semantic block masking—and using additive property descriptors to form Δ -supervision signals—effectively sensitizes the learned representations to property shifts induced by local structural perturbations. This leads to a marked reduction in the prediction error for downstream regression tasks.

Notably, even the MSHG-MAE_base variant outperformed most existing pretraining frameworks on regression tasks, achieving RMSEs of 1.010 on FreeSolv and 0.743 on ESOL, surpassing all baselines except Uni-Mol. Compared to the base version, the delta variant incorporating Δ -PropAlign delivered consistent gains across all three data sets. This trend corroborates the observation discussed later regarding the “enhanced consistency between structural and property distances in the embedding space”, providing further empirical evidence for the efficacy of Δ -PropAlign.

3.1.3. T-SNE Visualization. To intuitively inspect the graph-level embeddings learned by MSHG-MAE_delta, we visualized the test set representations using t-SNE,³³ colored by labels for two classification tasks and by prediction residuals for two regression tasks, as shown in Figure 3.

For the BBBP and BACE classification tasks, the labeled t-SNE projections show that positive and negative samples form several relatively compact clusters in the embedding space, with only limited mixing at the cluster boundaries. This pattern of intracluster homogeneity and intercluster heterogeneity is consistent with the high NH@5 scores (0.809 for BBBP and 0.748 for BACE), where NH@5 quantifies the probability that a sample shares its discretized activity label with the majority of its 5 nearest neighbors in the original embedding space. These observations indicate that the pretrained molecular representations can discriminate between activity classes even without any downstream classification heads. In addition, the trustworthiness scores of around 0.98 for both data sets suggest that the 2D t-SNE projections largely preserve the local neighborhood structure of the original high-dimensional space, minimizing projection-induced artifacts. For the ESOL and FreeSolv regression tasks, samples were colored by

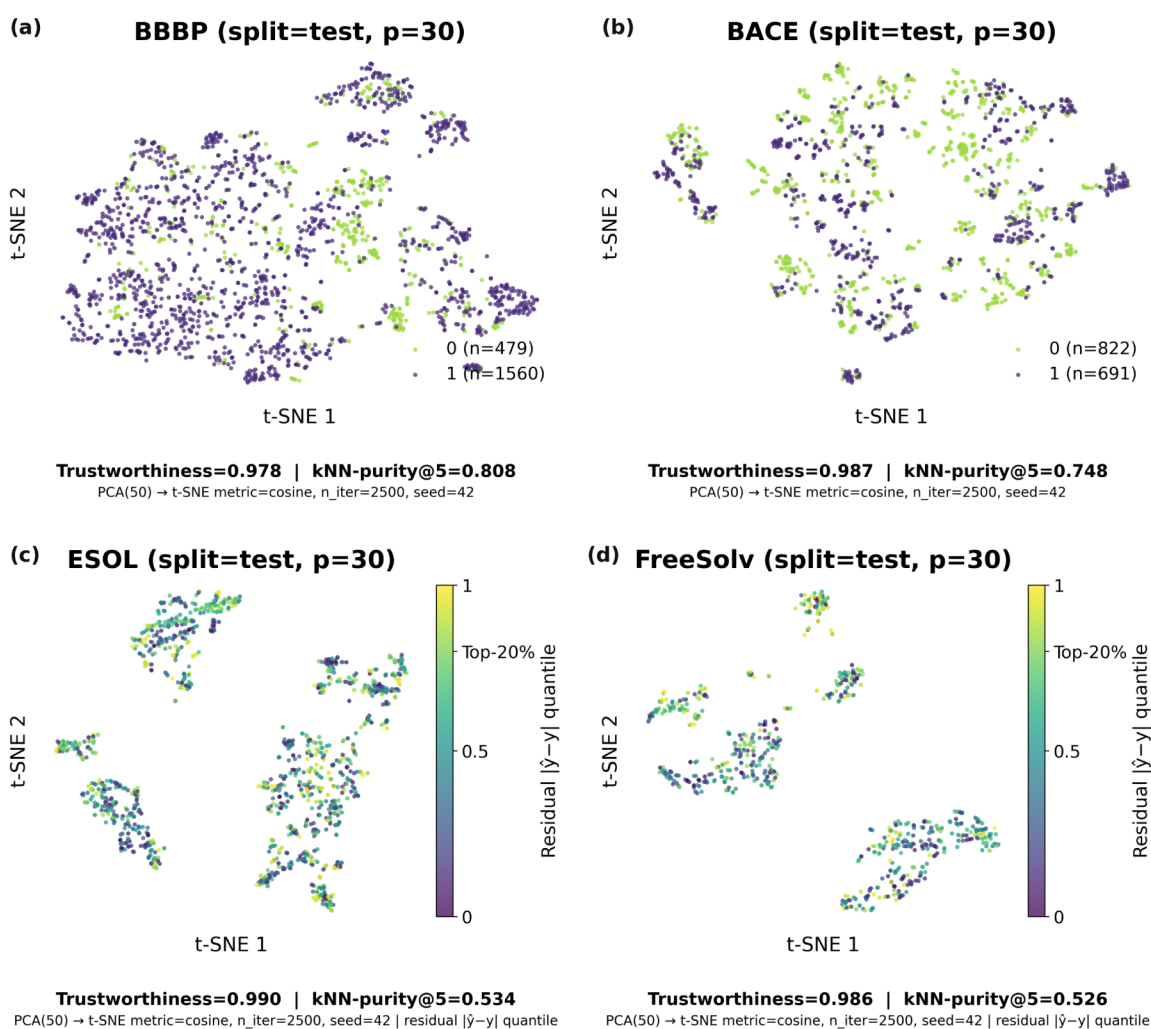


Figure 3. t-SNE visualization of pretrained molecular representations on classification (BBBP, BACE) and regression (ESOL, FreeSolv) benchmarks. Embeddings were extracted using a frozen encoder. (a, b) 2D projections colored by activity labels, showing clear separation between active and inactive classes. (c, d) Projections colored by prediction-residual quantiles; high-error samples concentrate near cluster boundaries. Trustworthiness:⁵⁴ preservation of each point's k -nearest neighbors from the original space (higher is better). $\text{NH}^{55}_{@5}$: neighborhood hit at $k = 5$ (fraction of samples whose 5 nearest neighbors share the same discretized label).

quantiles of their prediction residuals. We observed that the vast majority of samples exhibit lower-quantile residuals and are distributed within the interior of the embedding clusters. Conversely, samples with residuals in the top 20% quantile are predominantly located in intercluster regions or at cluster boundaries. This phenomenon is consistent with the empirical observation in regression tasks that “boundary samples are inherently more difficult to fit.”

3.2. Ablation Studies

3.2.1. Setup and Fairness. Unless otherwise specified, all ablation studies were conducted using scaffold splits across three random seeds. Computational budgets, training hyperparameters, and data processing protocols for pretraining and fine-tuning/linear probing were kept identical to those used for the full model. To preclude the possibility that performance gains stem merely from increased model capacity or parameter counts, we introduced two rigorous control baselines: Depth-matched (maintaining the architecture except for the multiscale convolution, but deepening the layers to match the total propagation steps of the multiscale approach) and Param-matched (maintaining the architecture except for the attention

mechanism, but increasing hidden dimensions/heads to match the parameter count of the full attention mechanism). All results are reported as mean $\pm$ standard deviation. The comparative results of the functional module ablations are illustrated in Figure 4.

3.2.2. Ablation of Individual Modules (Functional Modules and Hyperedge Types). A summary of the module/hyperedge ablations is provided in Table 6, and a 2×2 factorial ablation disentangling representation (Hypergraph vs CE) and masking (Semantic vs Random) are reported in Table 7.

3.2.2.1. Semantic Masking vs Random Masking. We replaced the “functional group-/ring-aware Semantic Block Masking” with standard random masking. Results show that semantic masking substantially improves regression (ESOL/FreeSolv) and yields comparable performance on BBBP under the hypergraph setting (Table 7). This suggests that random masking fails to force the model to focus on “high semantic density” regions, such as functional groups and rings. In contrast, semantic masking significantly enhances the quality of the reconstruction and cross-scale interaction modeling. To prevent information leakage through shortcut connections, we

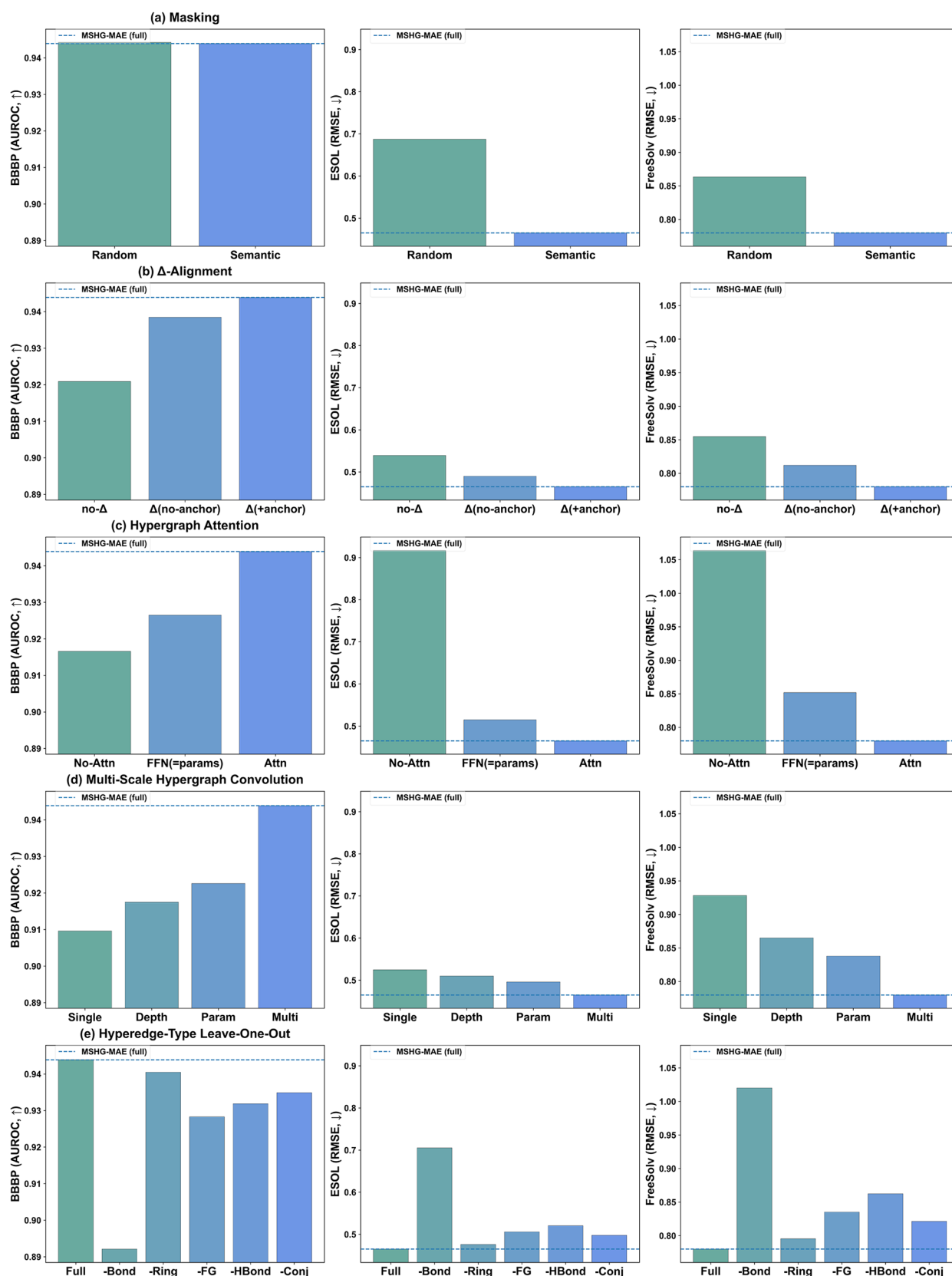


Figure 4. Overview of functional module ablation studies. Panels (a–e) summarize ablations on different components of the framework: (a) compares masking strategies (Random vs Semantic), (b) examines the Δ -PropAlign objective variants (no- Δ , Δ without anchor, and Δ with anchor), (c) evaluates hypergraph attention designs (No-Attn, FFN with matched parameters, and Attn), (d) tests multiscale hypergraph convolution settings (Single, Depth-matched, Param-matched, and Multi), and (e) reports leave-one-out results over hyperedge types (Full and removing Bond/Ring/FG/HBond/Conj). Results in each row are reported for BBBP (AUROC) and ESOL/FreeSolv (RMSE).

Table 6. Ablation of Mechanism Modules (Mean $\pm$ Std)^a

Configuration	BBBP (AUROC)	ESOL (RMSE)	FreeSolv (RMSE)
MSHG-MAE (Full)	0.9439 $\pm$ 0.02	0.4650 $\pm$ 0.05	0.7800 $\pm$ 0.11
w/o continued training	0.9209 $\pm$ 0.03	0.5394 $\pm$ 0.11	0.8549 $\pm$ 0.23
w/o multitype hyperedges	0.9050 $\pm$ 0.02	0.6770 $\pm$ 0.14	0.9744 $\pm$ 0.06
w/o multiscale convolution	0.9096 $\pm$ 0.03	0.5248 $\pm$ 0.06	0.9283 $\pm$ 0.14
w/o hypergraph attention	0.9166 $\pm$ 0.03	0.9170 $\pm$ 0.25	1.0638 $\pm$ 0.19
w/random masking	0.9442 $\pm$ 0.04	0.6874 $\pm$ 0.06	0.8635 $\pm$ 0.08
w/depth-matched	0.9175 $\pm$ 0.03	0.5100 $\pm$ 0.06	0.8650 $\pm$ 0.12
w/param-matched	0.9226 $\pm$ 0.03	0.5150 $\pm$ 0.06	0.8520 $\pm$ 0.12
w/FFN(=params)	0.9265 $\pm$ 0.03	0.5150 $\pm$ 0.06	0.8520 $\pm$ 0.12
w/ Δ (no-anchor)	0.9385 $\pm$ 0.03	0.4900 $\pm$ 0.05	0.8120 $\pm$ 0.11

^aNote: w/o means without, w/ means with.

simultaneously masked all hyperedges incident on the masked atoms.

We further study whether the hyperparameters in the semantic block masking strategy are tuned by chance. Specifically, we sweep the semantic to random masking mixture (100/0, 70/30, and 50/50), the overall MAE masking ratio (0.50, 0.60, 0.70, and 0.80), and the masking ratio annealing schedule in Δ -PropAlign (cosine, linear, constant). We report both reconstruction difficulty (best validation reconstruction loss and masked reconstruction MAE) and downstream performance over three seeds. Results show consistent trends and a stable performance region, with the chosen defaults close to the best trade-off across tasks (see [Supporting Information, Section S11](#)).

3.2.2.2. Δ -PropAlign. We compared three configurations: no- Δ -PropAlign, Δ -PropAlign (no anchor), and Δ -PropAlign (+anchor). Both Δ -PropAlign variants consistently improved upon the no- Δ -PropAlign baseline. Specifically, Δ -PropAlign (+anchor) proved the most robust on regression tasks (ESOL/FreeSolv) and yielded marginal gains on classification (BBBP). Referring to [Tables 3](#) and [7](#), simply removing the alignment step or replacing Δ -PropAlign with an equivalent number of standard self-supervised training steps resulted in significant degradation. This confirms that the performance gains stem from the consistency between “embedding differences and property differences” rather than merely from the extended training duration.

3.2.2.3. Hypergraph Attention Mechanism. To verify that improvements were not driven by increased capacity, we employed two controls: No-Attn (removing hypergraph attention while keeping other components unchanged) and FFN(=params) (substituting attention with a feed-forward network of equivalent parameter count, inserted after message

passing). The observed performance hierarchy was Attn > FFN(params) > No-Attn. This demonstrates that the improvement arises from the attribute-based gating and weighted aggregation rather than simple capacity expansion. Notably, on ESOL/FreeSolv, attention significantly reduced error, aligning with the chemical intuition that “different relation types and strengths make unequal contributions to solubility and solvation energy.”

3.2.2.4. Multi-Scale Hypergraph Convolution. The multi-scale approach applies parallel receptive fields corresponding to different diffusion orders of the incidence-based hypergraph operator introduced in [Section 2.3](#), all defined on the same underlying hypergraph. We compared this against three single-scale controls: single-scale (fixed receptive field), depth-matched (deeper single-scale matching propagation/FLOPs), and parameter-matched (wider single-scale matching parameter count). Multiscale consistently outperformed all single-scale variants across the three tasks. Even under “strictly aligned compute/parameter” conditions, the multiscale approach retained a significant advantage. This indicates that the gains derive not from “deeper or wider” networks but from cross-scale topological fusion capable of simultaneously capturing atomic neighborhoods, ring/conjugated substructures, and molecular-level dependencies.

3.2.2.5. Hypergraph vs Clique-Expansion (CE) Graph. From graph theory, given a hypergraph $\mathcal{H}$, we can transform it into a standard graph via Clique Expansion (CE), where all nodes within a hyperedge are connected pairwise to form a clique. For example, a hyperedge connecting nodes $\{A, B, C\}$ becomes the set of edges $\{(A, B), (A, C), (B, C)\}$ in the CE graph. We used CE to reduce high-order relations to binary edges—representing the upper limit of information expressible by standard graphs—to compensate for the inherent information deficit of simple graphs relative to hypergraphs using identical experimental configurations. Our results show that CE yields comparable performance on BBBP but is substantially worse on ESOL and FreeSolv, with the gap particularly pronounced in regression (ESOL and FreeSolv, [Table 7](#)). This suggests that clique expansion introduces redundant pairwise relations and may induce spectral distortion in the associated Laplacian operator, making it difficult to faithfully express multibody (high-order) co-occurrences and ultimately causing a performance gap in downstream tasks.^{56–58}

3.2.2.6. Contributions of Different Hyperedge Sources Assessed by Leave-One-Out (LOO) and Cumulative Addition. We conducted two complementary analyses to dissect the contributions of specific chemical relations ([Tables 7](#) and [8](#)). In the Leave-One-Out (LOO) analysis, we individually removed Bond, Ring, Functional Group (FG), Hydrogen Bond (HBond), or Conjugated Hyperedges from the full model ([Table 8](#)). In the Cumulative Addition analysis, we started from Bond hyperedges only and then successively added HBond, FG, Conj, and Ring hyperedges in this order ([Table 9](#)).

Table 7. Factorial Ablation of Representation $\times$ Masking (Mean $\pm$ Std)

Representation	Masking	BBBP (AUROC $\uparrow$)	ESOL (RMSE $\downarrow$)	FreeSolv (RMSE $\downarrow$)
Hypergraph	Random	0.9442 $\pm$ 0.0390	0.6874 $\pm$ 0.0584	0.8635 $\pm$ 0.0832
Hypergraph	Semantic	0.9439 $\pm$ 0.0207	0.4650 $\pm$ 0.0480	0.7800 $\pm$ 0.1110
CE	Random	0.9202 $\pm$ 0.0730	1.1138 $\pm$ 0.0540	1.2100 $\pm$ 0.0200
CE	Semantic	0.9423 $\pm$ 0.0104	0.9101 $\pm$ 0.2009	0.9558 $\pm$ 0.0633

Table 8. Ablation: Removing One Hyperedge Type (Mean $\pm$ Std)

Model (removed module)	BBBP (AUROC)	ESOL (RMSE)	FreeSolv (RMSE)
MSHG-MAE(Full)	0.9439 $\pm$ 0.02	0.4650 $\pm$ 0.05	0.7800 $\pm$ 0.11
w/o Bond	0.8921 $\pm$ 0.03	0.7053 $\pm$ 0.10	1.0200 $\pm$ 0.21
w/o Ring	0.9405 $\pm$ 0.02	0.4761 $\pm$ 0.05	0.7954 $\pm$ 0.10
w/o FG	0.9283 $\pm$ 0.03	0.5059 $\pm$ 0.06	0.8352 $\pm$ 0.12
w/o HBond	0.9319 $\pm$ 0.03	0.5204 $\pm$ 0.08	0.8626 $\pm$ 0.14
w/o Conj	0.9349 $\pm$ 0.03	0.4978 $\pm$ 0.06	0.8213 $\pm$ 0.12

Table 9. Cumulative Addition Hyperedge Types (Mean $\pm$ Std)

Configuration (added module)	BBBP (AUROC)	ESOL (RMSE)	FreeSolv (RMSE)
Bond only	0.9050 $\pm$ 0.02	0.6770 $\pm$ 0.14	0.9744 $\pm$ 0.06
Bond + HBond	0.9114 $\pm$ 0.02	0.6251 $\pm$ 0.11	0.9570 $\pm$ 0.06
Bond + HBond + FG	0.9352 $\pm$ 0.02	0.5236 $\pm$ 0.05	0.9059 $\pm$ 0.06
Bond + HBond + FG + Conj	0.9405 $\pm$ 0.02	0.4761 $\pm$ 0.05	0.7954 $\pm$ 0.10
All (Full)	0.9439 $\pm$ 0.02	0.4650 $\pm$ 0.05	0.7800 $\pm$ 0.11
CE (clique expansion)	0.9423 $\pm$ 0.01	0.9101 $\pm$ 0.20	0.9558 $\pm$ 0.06

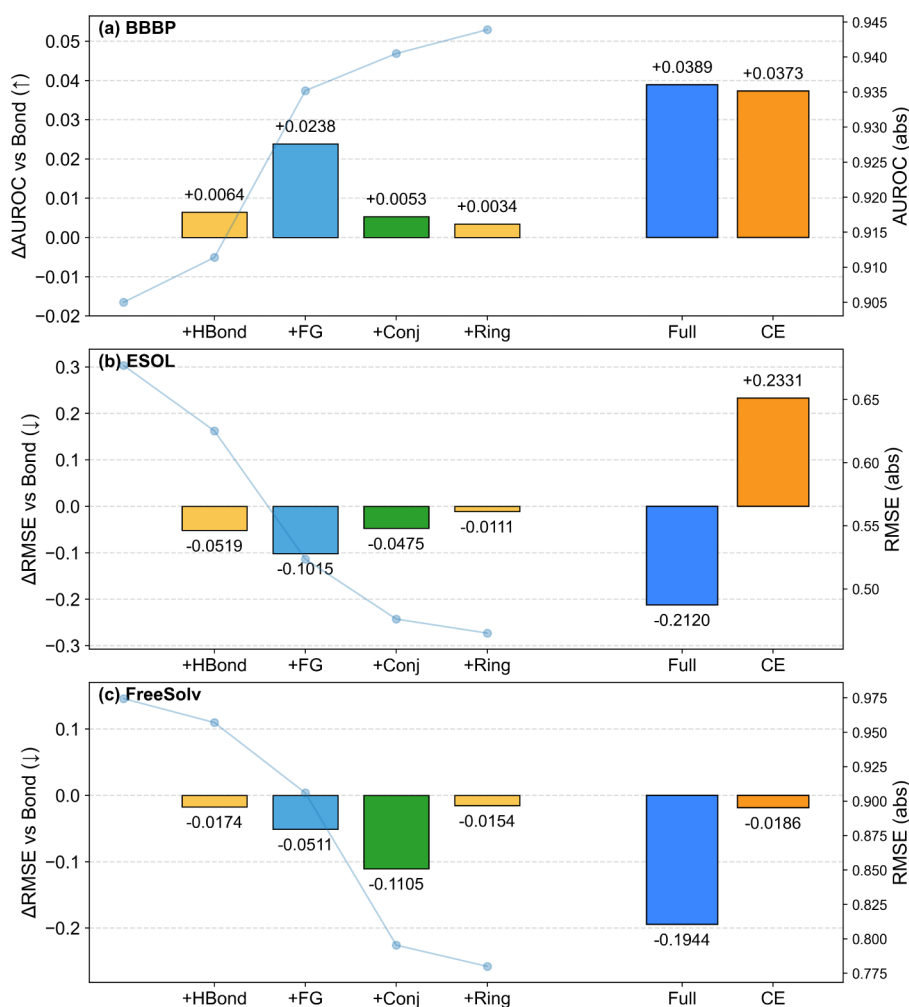


Figure 5. Stepwise contribution analysis of molecular-level features to model predictive performance. As hyperedge types are cumulatively added on top of the Bond-only baseline, panel (a) reports the relative improvement on BBBP (Δ AUROC), while panels (b) and (c) report the corresponding improvements on ESOL and FreeSolv (Δ RMSE), respectively. “Full” denotes the complete hypergraph model, and “CE” denotes the clique-expansion control.

Table 8 indicates that removing bond hyperedges causes the most severe performance degradation across all tasks (BBBP, ESOL, and FreeSolv), confirming that the topological information encoded by fundamental covalent bonds is

irreplaceable. The removal of HBond hyperedges had a particularly pronounced impact on ESOL and FreeSolv, aligning with the critical role of hydrogen bonding in determining the solubility and solvation energy. Removing

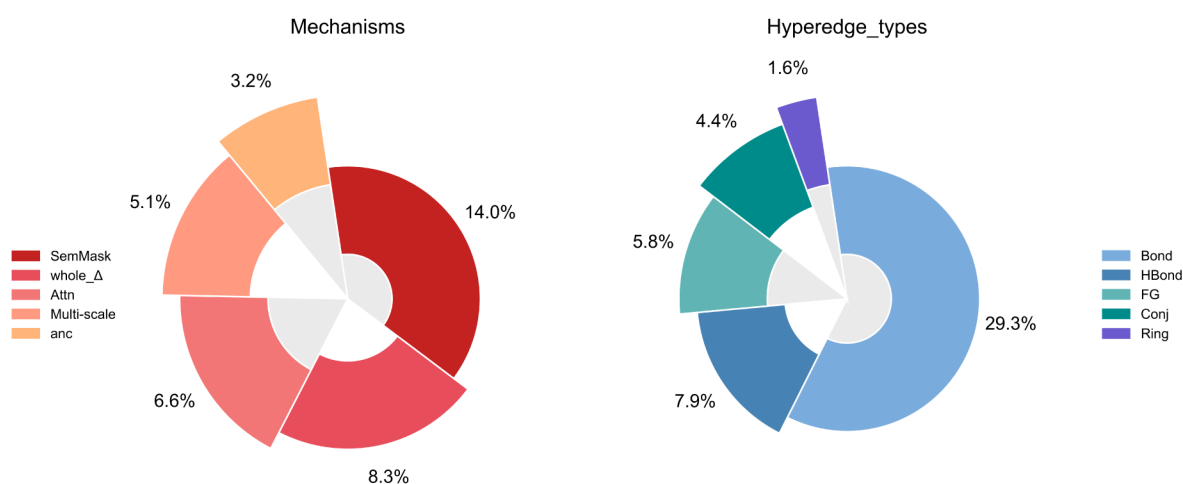


Figure 6. Nightingale rose diagram illustrates the relative contributions of different hyperedge types and functional modules. The radius of each sector represents the percentage improvement of the full model over the corresponding ablation baseline (e.g., without FG, without Attention) across three tasks. Larger radii indicate a stronger contribution to model performance. For classification (BBBP), we define the relative improvement as in eq 8.

the Ring, FG, or Conj hyperedges resulted in moderate degradation, suggesting that ring structures, functional groups, and conjugated systems provide complementary high-order semantic information.

As shown in Table 9, the cumulative addition experiment exhibited a monotonic performance improvement throughout the process, ultimately recovering the full model's performance. Notably, adding FG yielded the largest marginal gain on BBBP and ESOL, while adding Conj provided the most significant marginal boost for FreeSolv (Figure 5). These findings demonstrate the additivity and complementarity of multisource high-order relations in molecular representation.

Taken together, the LOO and cumulative addition results provide convergent evidence that the multisemantic hypergraph is not an incidental collection of relations, but a structured fusion of complementary chemical cues. Bond hyperedges serve as the indispensable covalent backbone, while HBond, FG, Conj, and Ring hyperedges capture noncovalent interactions and substructure-level semantics that yield consistent, task-dependent improvements when incorporated. This modular design also enables flexible pruning of relation types without changing the core architecture, which allows adapting the representation to different data sets or computational budgets.

3.2.3. Summary of Ablation Studies. The gains from the four modules—Semantic Masking, Δ -PropAlign, Hypergraph Attention, and Multiscale Convolution—are mutually independent and additive: removing any single module results in significant degradation on at least one data set. The relative contributions of each module are visualized in Figure 6. The full model achieves optimal performance across all three data sets simultaneously (highest BBBP-AUROC; lowest ESOL/FreeSolv-RMSE).

The Depth/Param-matched controls ruled out interpretations based on computational budget or capacity; the FFN(=params) control further eliminated the possibility that performance gains were solely due to increased parameter count. The CE control clearly demonstrated that explicit hypergraphs are more suitable than clique expansions for modeling multibody relations and predicting properties.

Consistent with the interpretability analysis in Section 3.3, Δ -PropAlign did not disrupt the geometric organization of structural/functional group semantics (equivalent Silhouette scores; improved Davies–Bouldin index). Instead, it further enhanced the consistency of embedding distance-semantic difference, which corresponds directly to the downstream gains observed in regression/classification tasks.

We further include a mask-conditioned additive control baseline (MaskPool), which explicitly provides the pooled representation of the masked subgraph to facilitate functional-group recognition and summation. To minimize confounds from parametrization or optimization, MaskPool and Δ -PropAlign use the same backbone configuration and training pipeline, and MaskPool differs only by an additional linear layer for concatenation-based Δ -Prediction (a negligible parameter increase). Moreover, following the additive-control design, we disabled the absolute anchor term for MaskPool. Under these matched settings, Δ -PropAlign still consistently outperforms MaskPool (Supporting Information, Section S11.1 and Table S7), which argues against a simple additive lookup explanation.

$$\Delta\text{AUROC} = \max(0, \text{AUROC}_{\text{ours}} - \text{AUROC}_{\text{ctrl}}) \quad (8)$$

For regression (ESOL/FreeSolv), we compute the percentage improvement as in eq 9:

$$\Delta\% = \max\left(0, \frac{\text{RMSE}_{\text{ctrl}} - \text{RMSE}_{\text{ours}}}{\text{RMSE}_{\text{ctrl}}}\right) \times 100\% \quad (9)$$

Finally, the values are averaged across the three data sets to derive the contribution of each module shown in the Figure.

3.3. Model Interpretability Analysis

Deep learning has demonstrated remarkable capabilities in complex pattern recognition and high-dimensional data modeling, showing significant advantages in tasks such as molecular property prediction. However, the “black box” nature of these models and the lack of interpretability in their internal representations continue to limit their widespread adoption in scientific research and critical applications. To enhance trust and understandability, increasing research attention is being directed toward the representation

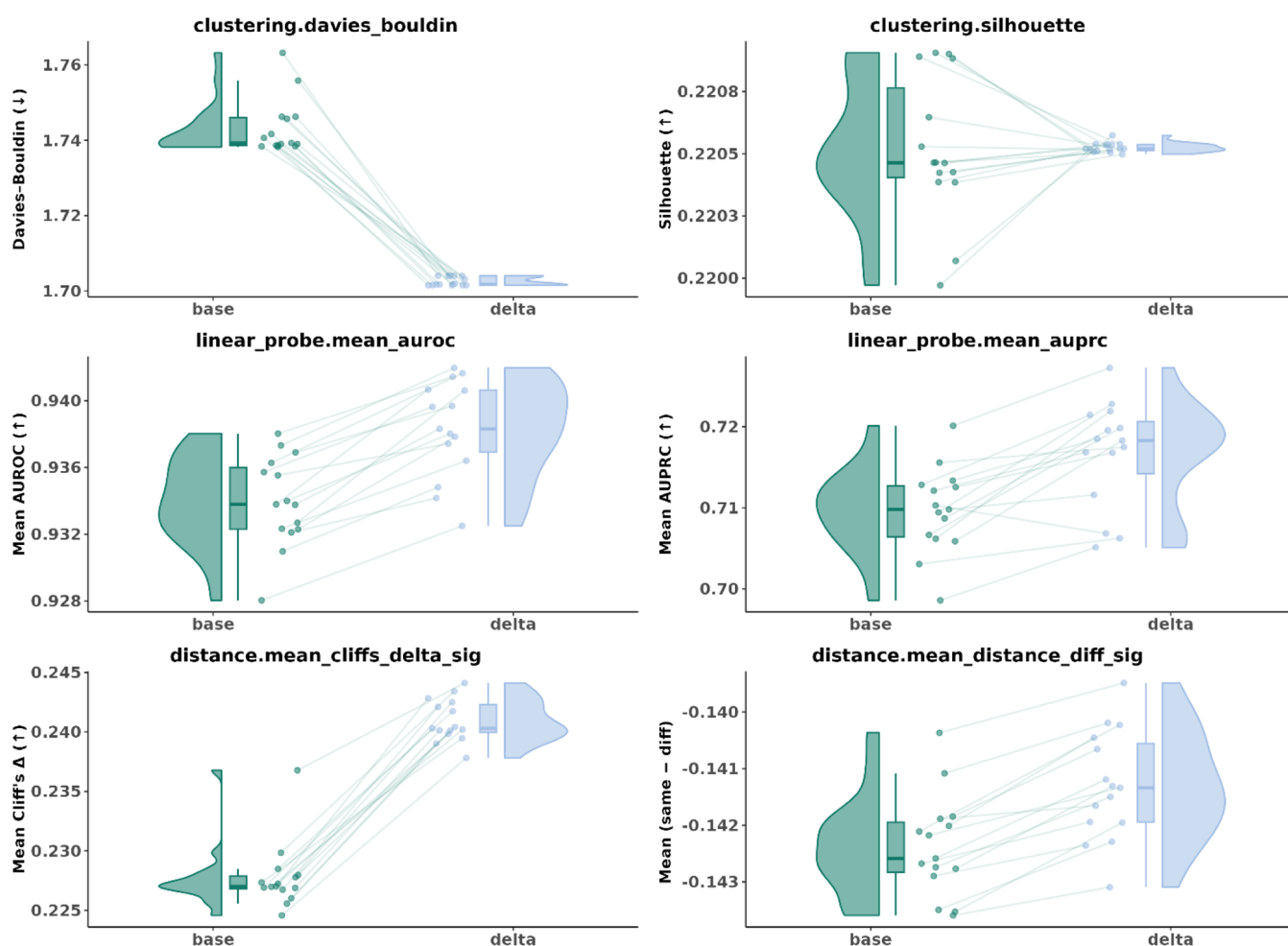


Figure 7. Robustness analysis of geometric and semantic consistency metrics across 15 random seeds. Pairwise comparisons (base vs delta) are performed on the BBBP data set (2,039 molecules). Violin plots display the distribution density, while box plots indicate the median and interquartile range (IQR). Lines connect data points from the same random seed to visualize directional trends. The vertical arrows (↑ or ↓) indicate the direction of better performance for each metric. The Δ -PropAlign model (delta) shows consistent improvements in compactness (Davies–Bouldin) and semantic distance consistency (Cliff's δ) without degrading structural semantics.

interpretability of model internals—specifically, whether the model has learned representations consistent with human-understandable chemical semantics (structure or functional groups).

In this work, we investigate the interpretability of Δ -PropAlign from a representation learning perspective, addressing two core questions: (a) Does Δ -PropAlign introduce property sensitivity without degrading the semantic encoding of structures/functional groups? (b) Is this improvement stable across random initializations and metrics? To this end, we compared the base model (base) with the Δ -PropAlign continued-training model (delta) on the BBBP data set (2,039 molecules) using the same data splits as in Section 3.1. Results were aggregated over 15 random seeds (reported as mean $\pm$ standard deviation). For cluster $\times$ functional group enrichment and intra/interclass distance analyses, we employed the one-sided Fisher's exact test and Mann–Whitney U test, respectively, controlling the False Discovery Rate (FDR) at $q = 0.05$ using the Benjamini–Hochberg procedure⁵⁹ (details in Supporting Information).

The analysis was designed to cover four complementary perspectives. First, Internal Clustering Metrics (Calinski–Harabasz,⁶⁰ Davies–Bouldin,⁶¹ Silhouette⁶²) were used to

quantify the structural separability and compactness of the embedding space. Second, a functional-group linear probe (24 classes) was trained on top of the frozen encoder to assess how easily chemical semantics can be decoded from the learned representations. Third, a distance–property/semantic consistency analysis was conducted to characterize the trade-off between property sensitivity and structural semantic fidelity: property consistency was evaluated on ESOL by correlating the pairwise embedding-distance matrix with the solubility-difference matrix, whereas semantic consistency was evaluated on BBBP by correlating embedding distances with functional-group differences. Finally, a cluster $\times$ functional-group enrichment analysis based on Fisher's exact test with Benjamini–Hochberg FDR control ($q = 0.05$) was performed to examine whether the learned clusters correspond to chemically interpretable groupings. Through these complementary analyses, we aim to determine whether Δ -PropAlign enhances representation sensitivity to molecular properties while preserving chemical semantic structure and whether this improvement remains robust across multiple random initializations and metrics. We additionally provide two supplementary analyses that directly address generality across task types: a linear probe correlation visualization on ESOL and an activity

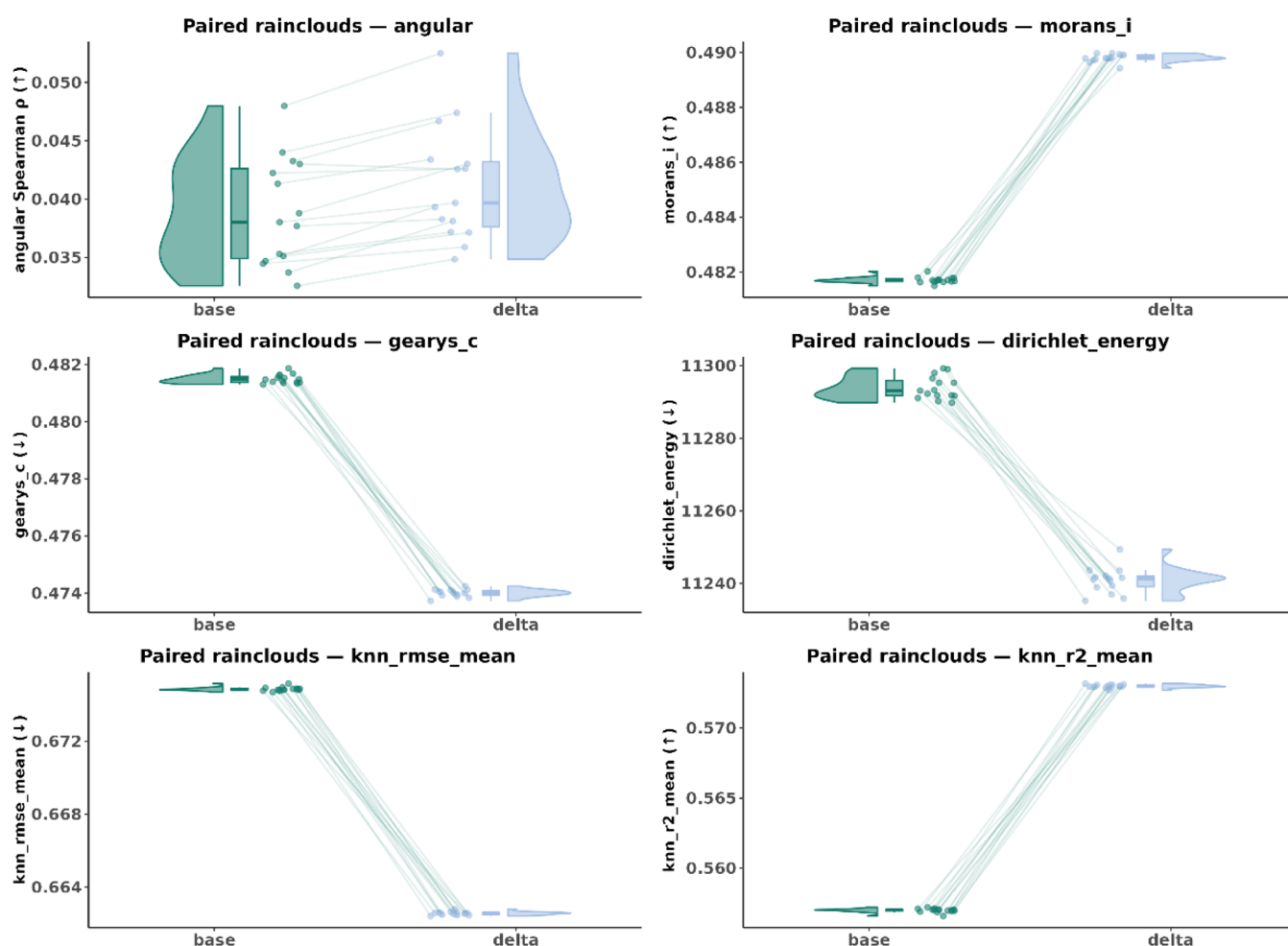


Figure 8. Quantitative assessment of distance–property alignment on the ESOL data set. Paired raincloud plots compare the base model (green) and the Δ -PropAlign model (delta, blue) across 15 independent random seeds. Six metrics (Angular Distance Spearman ρ , Moran's I, Geary's C, Dirichlet Energy, kNN-RMSE, and kNN-R²) were employed to evaluate the consistency between latent embedding distances and continuous property differences. Each plot combines a kernel density estimate (violin), a box plot indicating the median and interquartile ranges (IQR), and individual data points. Gray lines connect results from identical random seeds to highlight the consistent directional improvement induced by Δ -PropAlign. The vertical arrows ($\uparrow$ or $\downarrow$) next to the axis labels indicate the direction of better performance (e.g., higher is better for Spearman ρ , lower is better for Geary's C).

cliff sensitivity analysis on BBBP (Supporting Information, Sections S8.6–S8.7).

3.3.1. Preservation of Structural and Semantic Fidelity. To ensure that the introduction of property sensitivity does not degrade the model's understanding of chemical structures, we evaluated the geometric organization and linear separability of the learned representations based on Figure 7. Unsupervised clustering metrics on the BBBP data set indicate that the embedding structure remains stable after Δ -PropAlign training: the Calinski–Harabasz (CH) index decreased slightly ($\Delta = -27.56 \pm 0.067$), the Davies–Bouldin (DB) index improved (from 1.743 to 1.703, lower is better), and the Silhouette coefficient remained essentially unchanged. The number of clusters and the noise ratio were identical across models (5 clusters). These results suggest that the global geometric organization of structural semantics is preserved and, in some cases, slightly compacted.

We further assessed the explicit decodability of chemical semantics via a linear probe trained on top of the frozen encoder. For 24 functional-group probe tasks, the separability consistently improved: mean AUROC increased from 0.9340

to 0.9383 ($\Delta = +0.004$), and mean AUPRC rose from 0.7097 to 0.7167 ($\Delta = +0.007$). The low variance across random seeds indicates that functional-group identities became more linearly separable and stable. Finally, a cluster $\times$ functional-group enrichment analysis (one-sided Fisher's exact test with Benjamini–Hochberg FDR control at $q = 0.05$) showed that the number of significantly enriched terms remained nearly unchanged between the base and Δ -PropAlign models, confirming that the semantic interpretability of the clusters is not diminished by the injection of property-alignment signals.

3.3.2. Enhancement of Property Sensitivity. To measure whether semantic differences were more faithfully reflected in the representation geometry, we first examined the pairwise consistency between embedding distances and semantic differences: the number of significant pairs (after multiple comparison control) increased from 25 to 27. For these significant pairs, Cliff's δ rose from 0.2277 to 0.2409 ($\Delta = +0.013 \pm 0.002$), while the mean distance difference itself remained almost unchanged, confirming the enhanced pairwise consistency (see Figure 7).

Subsequently, we used six metrics—Angular Distance Spearman ρ ($\uparrow$), Moran's I^{63} ($\uparrow$), Geary's C^{64} ($\downarrow$), Dirichlet Energy^{65,66} (graph Laplacian quadratic form, $\downarrow$ = smoother), kNN-RMSE ($\downarrow$), and kNN- R^2 ($\uparrow$)—to evaluate the consistency between embedding distances and property differences on the ESOL data set, as summarized in Figure 8. Metric definitions and parameters are provided in Supporting Information (Section S8), with additional ESOL linear probe visualization and BBBP activity cliff analyses in Sections S8.6–S8.7.

Our results show that Spearman ρ increased, indicating that directional changes in the embedding space better reflect continuous property changes (i.e., the property gradient is more monotonic in the latent space). The increase in Moran's I and decreases in Geary's C and Dirichlet Energy collectively indicate that the property distribution on the kNN graph is smoother, and the local manifold is more continuous. Simultaneously, the decrease in kNN-RMSE and increase in kNN- R^2 demonstrate lower prediction error and higher explainability within local neighborhoods. Overall, Δ -PropAlign makes the latent space response to continuous properties smoother and more directionally consistent while maintaining a stable semantic structure, which corresponds to the performance gains observed on regression benchmarks.

To complement the distance-based ESOL metrics, we also train a linear probe with frozen encoders and visualize predicted versus true solubility values. The Δ -PropAlign model shows closer agreement to the $y = x$ diagonal and higher Pearson correlation, which provides a direct view of improved global consistency (Supporting Information, Section S8.6, Figure S1).

More importantly, to test whether the representation improvement extends to challenging classification regimes, we conduct an activity cliff analysis on BBBP. We construct test set pairs with high structural similarity but opposite labels and then evaluate them using a linear probe with frozen encoders. Δ -PropAlign achieves higher pair consistency and larger $|\Delta\text{Pred}|$, indicating sharper sensitivity to subtle structural edits that flip biological outcomes (Supporting Information, Section S8.7, Figure S2).

3.4. Summary of Findings

Overall, the findings indicate that structural and functional-group semantics are preserved or slightly improved: the geometric organization of the embedding space is noninferior, with the Silhouette coefficient remaining unchanged, while compactness improves, as reflected by a lower Davies–Bouldin index. Functional-group separability also exhibits a modest but consistent gain, with increases in both AUROC and AUPRC under a controlled aggregation variance. In terms of semantic distance consistency on BBBP, the number of significant pairs and Cliff's δ both increase, and, on the ESOL data set, improvements in Spearman's ρ , Moran's I , Geary's C , Dirichlet energy, and kNN-based metrics collectively indicate that the embedding geometry aligns more closely with solubility differences.

These phenomena align with our methodological design: Semantic Block Masking allows the model to learn “meaningful directions of structural perturbation” in high-semantic-density regions (functional groups/rings), while Δ -PropAlign anchors the “pre/post-masking” embedding difference to the property difference. This enhances semantic distance consistency without sacrificing the separability of functional group

identities. In practice, Δ -PropAlign is preferred when tasks are property-oriented or data-limited; when structural pattern recognition is paramount, one may reduce the Δ -PropAlign weight or adopt a staged training strategy (“structure-first, then Δ -PropAlign”) to maintain structural semantic equivalence while gaining property sensitivity.

4. DISCUSSION

The contributions of this work are threefold. First, we propose MSHG-MAE, a hypergraph masked autoencoder specifically designed for molecular systems, which models higher-order chemical semantics such as functional groups, rings, conjugation, and hydrogen bonds as hyperedges and performs semantic block masking with joint reconstruction on both nodes and hyperedges. Second, we introduce Δ -PropAlign as a weakly supervised continued training phase following MSHG-MAE pretraining: for each molecule, we perform one forward pass on the original graph and another with semantic blocks masked and explicitly align the difference in graph-level predictions with the property change computed from additive RDKit descriptors over the masked region, thereby encoding the relationship between structural modification and property change into training and enhancing both predictive accuracy and representation interpretability. Third, we conduct systematic, large-scale ablation studies over hyperedge types and key functional modules, elucidating the individual contributions of each hyperedge type and architectural component to the molecular property prediction.

From a representation-learning perspective, the main contribution of the proposed hypergraph masked autoencoder is not merely that it “uses more edges” but that it changes what the encoder is allowed to treat as a primitive unit of information. By elevating rings, functional groups, hydrogen bond patterns, and conjugated systems to first-class hyperedges, the model no longer has to reconstruct such motifs indirectly from chains of pairwise bonds but can instead learn over a vocabulary of chemically meaningful semantic blocks. The semantic-block masking objective then imposes a stringent information bottleneck: to succeed, the encoder must infer the missing hyperedges and their attributes from the remaining multiscale context, effectively aligning atom-level representations with motif-level structure in a self-consistent way. This differs qualitatively from contrastive or nodewise masking schemes, which encourage global invariance but often underconstrain how higher-order chemistry is organized in the latent space. The empirical gains we observe, particularly on regression benchmarks, suggest that this inductive bias steers the model toward embeddings in which local perturbations at the hyperedge level correspond more smoothly to changes in physicochemical properties, a prerequisite for label-efficient prediction.

Synthesizing the above results, Δ -PropAlign can be viewed as a form of “Property-Aware Geometric Calibration.” Starting from a semantically well-structured base representation, finite-step Δ -tuning explicitly encodes the directional gradients of continuous properties into the latent space. This makes the local manifold smoother and more monotonic with respect to property changes while maintaining or slightly enhancing the original functional group structure. This mechanism explains the small but stable improvements in distance–property/semantic consistency metrics and aligns with the performance gains in downstream tasks (e.g., ESOL). Crucially, the model does not achieve performance by drastically rewriting the

representation (“trading” structure for property) but by injecting a geometric layer aligned with the property manifold on top of the base representation. Note that this alignment trains the model’s sensitivity—learning the physical law that “local structural changes in chemically active regions can induce drastic property shifts”—rather than merely learning to calculate properties like LogP.

Despite the practical benefits and interpretability gains brought by MSHG-MAE and Δ -PropAlign, the current design still has several limitations. First, Δ -PropAlign relies on additive contributions from RDKit, which inevitably introduces proxy bias and may limit generalization on nonphysicochemical properties (e.g., toxicity, biological targets). To address this, future work could introduce weak supervision from real-world matrix molecular pairs (MMPs), transcending rule-based proxies. By curating pairs from large databases (e.g., ChEMBL) that differ by a well-defined local semantic modification and have annotated experimental activity differences, we could train the Δ -PropAlign module to learn real Δ -Properties directly from data.

Second, regarding the construction of hydrogen bond hyperedges, we utilized lightweight 3D conformer generation (ETKDG + MMFF) rather than relying on 2D SMARTS patterns. While this introduces a preprocessing dependency, our ablation studies confirm that the geometric semantics provided by explicit HBonds significantly contribute to performance. Future work could explore dual-view architectures that learn to infer these 3D semantic motifs directly from 2D graphs, potentially removing the need for explicit conformer generation at the inference time in real-world deployments. Third, the hypergraph construction relies on heuristic rules for extracting structural primitives. Future work will explore learning the hyperedge vocabulary jointly with the encoder or adaptively adjusting it to reduce the dependence on manual chemical priors.

5. CONCLUSION

This work presents MSHG-MAE, a molecular pretraining framework integrating multiscale hypergraph convolution, semantic block masking, and Δ -PropAlign. Under a unified benchmark protocol with shared data splits and comparable fine-tuning budgets, MSHG-MAE matches or surpasses strong baselines on multiple regression tasks (e.g., ESOL, FreeSolv) while remaining competitive on classification tasks. Systematic interpretability analysis reveals that, in 15 independent trials on the BBBP data set, Δ -PropAlign enhances sensitivity to semantic differences and stabilizes linear separability without degrading structural/functional group semantics. Specifically, we observed equivalent Silhouette scores, improved Davies–Bouldin indices, and consistent increases in mean-AUROC/AUPRC and “semantic distance consistency”.

In summary, the two-stage framework from MSHG-MAE to Δ -PropAlign achieves robust gains under a unified protocol. Δ -PropAlign explicitly injects the “structural perturbation–property difference” relationship into training, improving representation interpretability. We have open-sourced the code, with the hope of providing a reproducible empirical foundation for future molecular representation and property modeling. Methodologically, this offers a path balancing performance and interpretability: the encoder not only “recognizes” structure but also “responds” to property changes induced by local semantic modifications. We believe that the training paradigm centering on structure–semantics–property

differences, combined with high-quality hypergraph masked autoencoders, will become a vital bridge connecting self-supervised representation learning with practical molecular design.

■ ASSOCIATED CONTENT

Data Availability Statement

The core code for MSHG-MAE is freely available on GitHub at: <https://github.com/Irzos/MSHG-MAE>. The pretraining corpus was sampled from the ZINC20 database (<http://zinc20.docking.org/>). The nine benchmark data sets used for downstream tasks (BBBP, BACE, ClinTox, HIV, SIDER, Tox21, ESOL, FreeSolv, and Lipophilicity) were obtained from MoleculeNet.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c02994>.

Detailed hyperedge construction rules and feature definitions for all semantic hyperedge types (Section S1), a matrix-form derivation of the multi-receptive-field hypergraph convolution operator and normalization scheme (Section S2), the SMARTS pattern library used for functional-group and motif identification (Section S3), a complete list of atomic features and hyperedge attributes (Section S4), pretraining corpus curation and standardization procedures based on ZINC20 (Section S5), implementation and hyperparameter details for pretraining and fine-tuning, including the Target Contribution Controller (TCC), the Δ -PropAlign schedule, and test-time postprocessing settings (Section S6), hardware configurations and training time statistics (Section S7), definitions and configurations of the representation-level evaluation metrics used in the interpretability analyses (Section S8), and detailed descriptions of the nine downstream benchmark data sets (Section S9), hyperedge statistics across data sets (Section S10), and additional ablation experiments (Section S11) (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Xian Wei – Department of Pharmacy, Youjiang Medical University for Nationalities, Baise 533000, China; Email: wei700@126.com

Yanmin Zhang – Laboratory of Molecular Design and Drug Discovery, School of Science, China Pharmaceutical University, Nanjing 211198, China; orcid.org/0000-0002-4075-7556; Email: yanminzhang@cpu.edu.cn

Authors

Ziyan Zhu – Laboratory of Molecular Design and Drug Discovery, School of Science, China Pharmaceutical University, Nanjing 211198, China

Yijie Wang – Laboratory of Molecular Design and Drug Discovery, School of Science, China Pharmaceutical University, Nanjing 211198, China

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.jcim.5c02994>

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was financially supported by the Natural Science Foundation of Jiangsu Province (no. BK20242073), the Natural Science Foundation of Guangxi Province (no. 2025GXNSFAA069993), the Fundamental Research Funds for the Central Universities (no. 2632025ZD04), the Philosophy and Social Science Research in Colleges and Universities in Jiangsu Province (no. 2023SJZD130, no. 2024SJYB0082), and the Key-funded Research of Jiangsu Provincial Excellent Applied Social Science Project (no.25SYA-035).

ABBREVIATIONS

AUPRC, area under the precision–recall curve; AUROC, area under the receiver operating characteristic curve; BACE, β -site amyloid precursor protein cleaving enzyme 1 inhibition data set; BBBP, blood–brain barrier penetration data set; ClinTox, clinical toxicity data set; D-MPNN, directed message passing neural network; ECFP, extended-connectivity fingerprint; ESOL, aqueous solubility benchmark data set; FG, functional group; FreeSolv, hydration free energy benchmark data set; GNN, graph neural network; HBond, hydrogen bond Hyperedge; HIV, human immunodeficiency virus data set; Lipo, lipophilicity benchmark data set; MAE, masked autoencoder; MMP, matched molecular pair; MSHG-MAE, multiscale hypergraph masked autoencoder; RMSE, root-mean-square error; SIDER, side effect resource toxicity data set; SMILES, simplified molecular-input line-entry system; SSL, self-supervised learning; TCC, target contribution controller; Tox21, toxicology in the 21st century data set; TPSA, topological polar surface area; Uni-Mol, universal three-dimensional molecular representation learning framework; WSD, warmup–stable–decay learning rate scheduler; Δ -PropAlign, Δ -Property Alignment.

REFERENCES

- (1) Brown, D. G.; Wobst, H. J.; Kapoor, A.; Kenna, L. A.; Southall, N. Clinical Development Times for Innovative Drugs. *Nat. Rev. Drug Discovery* **2022**, *21* (11), 793–794.
- (2) Wouters, O. J.; McKee, M.; Luyten, J. Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009–2018. *JAMA* **2020**, *323* (9), 844–853.
- (3) Li, Z.; Jiang, M.; Wang, S.; Zhang, S. Deep Learning Methods for Molecular Representation and Property Prediction. *Drug Discovery Today* **2022**, *27* (12), 103373.
- (4) Yang, K.; Swanson, K.; Jin, W.; Coley, C.; Eiden, P.; Gao, H.; Guzman-Perez, A.; Hopper, T.; Kelley, B.; Mathea, M.; Palmer, A.; Settels, V.; Jaakkola, T.; Jensen, K.; Barzilay, R. Analyzing Learned Molecular Representations for Property Prediction. *J. Chem. Inf. Model.* **2019**, *59* (8), 3370–3388.
- (5) Chuang, K. V.; Gunsalus, L. M.; Keiser, M. J. Learning Molecular Representations for Medicinal Chemistry: Miniperspective. *J. Med. Chem.* **2020**, *63* (16), 8705–8722.
- (6) David, L.; Thakkar, A.; Mercado, R.; Engkvist, O. Molecular Representations in AI-Driven Drug Discovery: A Review and Practical Guide. *J. Cheminf.* **2020**, *12* (1), 56.
- (7) Rogers, D.; Hahn, M. Extended-Connectivity Fingerprints. *J. Chem. Inf. Model.* **2010**, *50* (5), 742–754.
- (8) Jiang, D.; Wu, Z.; Hsieh, C.-Y.; Chen, G.; Liao, B.; Wang, Z.; Shen, C.; Cao, D.; Wu, J.; Hou, T. Could Graph Neural Networks Learn Better Molecular Representation for Drug Discovery? A Comparison Study of Descriptor-Based and Graph-Based Models. *J. Cheminf.* **2021**, *13* (1), 12.
- (9) Xiong, Z.; Wang, D.; Liu, X.; Zhong, F.; Wan, X.; Li, X.; Li, Z.; Luo, X.; Chen, K.; Jiang, H.; Zheng, M. Pushing the Boundaries of Molecular Representation for Drug Discovery with the Graph Attention Mechanism. *J. Med. Chem.* **2020**, *63* (16), 8749–8760.
- (10) Heid, E.; Greenman, K. P.; Chung, Y.; Li, S.-C.; Graff, D. E.; Vermeire, F. H.; Wu, H.; Green, W. H.; McGill, C. J. Chemprop: A Machine Learning Package for Chemical Property Prediction. *J. Chem. Inf. Model.* **2024**, *64* (1), 9–17.
- (11) Wang, Y.; Wang, J.; Cao, Z.; Barati Farimani, A. Molecular Contrastive Learning of Representations via Graph Neural Networks. *Nat. Mach. Intell.* **2022**, *4* (3), 279–287.
- (12) Choi, J.; Nam, G.; Choi, J.; Jung, Y. A Perspective on Foundation Models in Chemistry. *JACS Au* **2025**, *5* (4), 1499–1518.
- (13) Liu, S.; Wang, H.; Liu, W.; Lasenby, J.; Guo, H.; Tang, J. Pre-Training Molecular Graph Representation with 3D Geometry. *arXiv* **2021**, DOI: 10.48550/arXiv.2110.07728.
- (14) He, K.; Chen, X.; Xie, S.; Li, Y.; Dollar, P.; Girshick, R. Masked Autoencoders Are Scalable Vision Learners. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*; IEEE: New Orleans, LA, USA, 2022; pp 15979–15988.
- (15) Hou, Z.; Liu, X.; Cen, Y.; Dong, Y.; Yang, H.; Wang, C.; Tang, J. GraphMAE: Self-Supervised Masked Graph Autoencoders. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining; KDD'22*; Association for Computing Machinery: New York, NY, USA, 2022; pp 594–604.
- (16) Hou, Z.; He, Y.; Cen, Y.; Liu, X.; Dong, Y.; Kharlamov, E.; Tang, J. GraphMAE2: A Decoding-Enhanced Masked Self-Supervised Graph Learner. *Proceedings of the ACM Web Conference 2023*; ACM: Austin, TX, USA, 2023, 737–746.
- (17) Chithrananda, S.; Grand, G.; Ramsundar, B. ChemBERTa: Large-Scale Self-Supervised Pretraining for Molecular Property Prediction. *arXiv* **2020**, DOI: 10.48550/arXiv.2010.09885.
- (18) Ahmad, W.; Simon, E.; Chithrananda, S.; Grand, G.; Ramsundar, B. ChemBERTa-2: Towards Chemical Foundation Models. *arXiv* **2022**, DOI: 10.48550/arXiv.2209.01712.
- (19) Xia, J.; Zhao, C.; Hu, B.; Gao, Z.; Tan, C.; Liu, Y.; Li, S.; Li, S. Z. Mole-BERT: Rethinking Pre-Training Graph Neural Networks for Molecules. In *The Eleventh International Conference on Learning Representations; ICLR*, 2023.
- (20) Rong, Y.; Bian, Y.; Xu, T.; Xie, W.; Wei, Y.; Huang, W.; Huang, J. Self-Supervised Graph Transformer on Large-Scale Molecular Data. In *Advances in Neural Information Processing Systems*; Curran Associates, Inc., 2020; Vol. 33, pp. 12559–12571.
- (21) Battiston, F.; Cencetti, G.; Iacopini, I.; Latora, V.; Lucas, M.; Patania, A.; Young, J.-G.; Petri, G. Networks beyond Pairwise Interactions: Structure and Dynamics. *Phys. Rep.* **2020**, *874*, 1–92.
- (22) Murgas, K. A.; Saucan, E.; Sandhu, R. Hypergraph Geometry Reflects Higher-Order Dynamics in Protein Interaction Networks. *Sci. Rep.* **2022**, *12* (1), 20879.
- (23) Mao, Z.; Hu, C.-S.; Li, J.; Liang, C.; Das, D.; Sumita, M.; Xia, K.; Tsuda, K. Molecule Graph Networks with Many-Body Equivariant Interactions. *J. Chem. Theory Comput.* **2025**, *21* (16), 7954–7966.
- (24) Chen, J.; Schwaller, P. Molecular Hypergraph Neural Networks. *J. Chem. Phys.* **2024**, *160*, 144307.
- (25) Wang, B.; Li, J.; Zhou, D.; Li, L.; Li, J.; Wang, E.; Hao, J.; Shi, L.; Lu, C.; Qiu, J.; Hou, T.; Cao, D.; Chen, G.; Heng, P. A Unified and Explainable Molecular Representation Learning for Imperfectly Annotated Data from the Hypergraph View. *Nat. Commun.* **2025**, *16* (1), 8717.
- (26) Du, W.; Zhang, S.; Cai, Z.; Li, X.; Liu, Z.; Fang, J.; Wang, J.; Wang, X.; Wang, Y. Molecular Merged Hypergraph Neural Network

for Explainable Solvation Gibbs Free Energy Prediction. *Research* **2025**, 8, 0740.

- (27) Lin, W.; Fung, C. C. A. Rethinking the Masking Strategy for Pretraining Molecular Graphs from a Data-Centric View. *ACS Omega* **2024**, 9 (19), 20832–20838.
- (28) Peng, T.; Li, Y.; Li, X.; Bian, J.; Xie, Z.; Sui, N.; Mumtaz, S.; Xu, Y.; Kong, L.; Xiong, H. Pre-Trained Molecular Language Models with Random Functional Group Masking. *Npj Artif. Intell.* **2025**, 1 (1), 28.
- (29) Fang, Y.; Zhang, Q.; Zhang, N.; Chen, Z.; Zhuang, X.; Shao, X.; Fan, X.; Chen, H. Knowledge Graph-Enhanced Molecular Contrastive Learning with Functional Prompt. *Nat. Mach. Intell.* **2023**, 5 (5), 542–553.
- (30) Gim, M.; Park, J.; Park, S.; Lee, S.; Baek, S.; Lee, J.; Nguyen, N.-Q.; Kang, J. MolPLA: A Molecular Pretraining Framework for Learning Cores, R-Groups and Their Linker Joints. *Bioinformatics* **2024**, 40 (Supplement_1), i369–i380.
- (31) To, V.-T.; Van Nguyen, P.-C.; Truong, G.-B.; Phan, T.-M.; Phan, T.-L.; Fagerberg, R.; Stadler, P. F.; Truong, T. N. K. Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions. *J. Chem. Inf. Model.* **2025**, 65 (18), 9443–9458.
- (32) Landrum, G. RDKit: Open-Source Cheminformatics Software; <http://www.rdkit.org>.
- (33) Weininger, D. SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules. *J. Chem. Inf. Comput. Sci.* **1988**, 28 (1), 31–36.
- (34) Downs, G. M.; Gillet, V. J.; Holliday, J. D.; Lynch, M. F. Review of Ring Perception Algorithms for Chemical Graphs. *J. Chem. Inf. Comput. Sci.* **1989**, 29 (3), 172–187.
- (35) Horton, J. D. A Polynomial-Time Algorithm to Find the Shortest Cycle Basis of a Graph. *SIAM J. Comput.* **1987**, 16 (2), 358–366.
- (36) Halgren, T. A. Merck Molecular Force Field. I. Basis, Form, Scope, Parameterization, and Performance of MMFF94. *J. Comput. Chem.* **1996**, 17 (5–6), 490–519.
- (37) Baker, E. N.; Hubbard, R. E. Hydrogen Bonding in Globular Proteins. *Prog. Biophys. Mol. Biol.* **1984**, 44 (2), 97–179.
- (38) McDonald, I. K.; Thornton, J. M. Satisfying Hydrogen Bonding Potential in Proteins. *J. Mol. Biol.* **1994**, 238 (5), 777–793.
- (39) Chen, Z.; Badrinarayanan, V.; Lee, C.-Y.; Rabinovich, A. GradNorm: Gradient Normalization for Adaptive Loss Balancing in Deep Multitask Networks. *Proceedings of the 35th International Conference on Machine Learning*; PMLR, 2018, 794–803.
- (40) Liu, S.; Johns, E.; Davison, A. J. End-To-End Multi-Task Learning With Attention. *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*; IEEE: Long Beach, CA, USA, 2019, 1871–1880.
- (41) Loshchilov, I.; Hutter, F. Decoupled Weight Decay Regularization. *arXiv*. **2018**, DOI: 10.48550/arXiv.1711.05101.
- (42) Hu, S.; Tu, Y.; Han, X.; He, C.; Cui, G.; Long, X.; Zheng, Z.; Fang, Y.; Huang, Y.; Zhao, W., et al. MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies. In *Proceedings of the Conference on Language Modeling (COLM 2024)*; COLM, 2024.
- (43) Wu, Z.; Ramsundar, B.; Feinberg, E. N.; Gomes, J.; Geniesse, C.; Pappu, A. S.; Leswing, K.; Pande, V. MoleculeNet: A Benchmark for Molecular Machine Learning. *Chem. Sci.* **2018**, 9 (2), 513–530.
- (44) Breiman, L. Random Forests. *Mach. Learn.* **2001**, 45 (1), 5–32.
- (45) Xu, K.; Hu, W.; Leskovec, J.; Jegelka, S. How Powerful Are Graph Neural Networks? In *International Conference on Learning Representations*; ICLR, 2018.
- (46) Kipf, T. N.; Welling, M. Semi-Supervised Classification with Graph Convolutional Networks. In *International Conference on Learning Representations*; ICLR, 2017.
- (47) Zhou, G.; Gao, Z.; Ding, Q.; Zheng, H.; Xu, H.; Wei, Z.; Zhang, L.; Ke, G. Uni-Mol: A Universal 3D Molecular Representation Learning Framework. In *International Conference on Learning Representations*; ICLR, 2022.
- (48) Zeng, X.; Xiang, H.; Yu, L.; Wang, J.; Li, K.; Nussinov, R.; Cheng, F. Accurate Prediction of Molecular Properties and Drug Targets Using a Self-Supervised Image Representation Learning Framework. *Nat. Mach. Intell.* **2022**, 4 (11), 1004–1016.
- (49) Irwin, J. J.; Tang, K. G.; Young, J.; Dandarchuluun, C.; Wong, B. R.; Khurelbaatar, M.; Moroz, Y. S.; Mayfield, J.; Sayle, R. A. ZINC20—A Free Ultralarge-Scale Chemical Database for Ligand Discovery. *J. Chem. Inf. Model.* **2020**, 60 (12), 6065–6073.
- (50) Vitter, J. S. Random Sampling with a Reservoir. *ACM Trans. Math. Softw.* **1985**, 11 (1), 37–57.
- (51) Heller, S.; McNaught, A.; Stein, S.; Tchekhovskoi, D.; Pletnev, I. InChI - the Worldwide Chemical Structure Identifier Standard. *J. Cheminf.* **2013**, 5 (1), 7.
- (52) Cliff, N. Dominance Statistics: Ordinal Analyses to Answer Ordinal Questions. *Psychol. Bull.* **1993**, 114 (3), 494–509.
- (53) Van der Maaten, L.; Hinton, G. Visualizing Data Using T-SNE. *J. Mach. Learn. Res.* **2008**, 9 (86), 2579–2605.
- (54) Venna, J.; Kaski, S. Neighborhood Preservation in Nonlinear Projection Methods: An Experimental Study. In *Artificial Neural Networks - ICANN 2001*; Dorffner, G.; Bischof, H.; Hornik, K.; Eds.; Springer: Berlin, Heidelberg, 2001; pp. 485–491.
- (55) Paulovich, F. V.; Nonato, L. G.; Minghim, R.; Levkowitz, H. Least Square Projection: A Fast High-Precision Multidimensional Projection Technique and Its Application to Document Mapping. *IEEE Trans. Vis. Comput. Graph.* **2008**, 14 (3), 564–575.
- (56) Mulas, R.; Kuehn, C.; Böhle, T.; Jost, J. Random Walks and Laplacians on Hypergraphs: When Do They Match? *Discrete Appl. Math.* **2022**, 317, 26–41.
- (57) Li, P.; Milenkovic, O. Submodular Hypergraphs: P-Laplacians, Cheeger Inequalities and Spectral Clustering. *Proceedings of the 35th International Conference on Machine Learning*; PMLR, 2018, 3014–3023.
- (58) Wang, F.; Pena-Pena, K.; Qian, W.; Arce, G. R. T-HyperGNNs: Hypergraph Neural Networks via Tensor Representations. *IEEE Trans. Neural Netw. Learn. Syst.* **2025**, 36 (3), 5044–5058.
- (59) Benjamini, Y.; Hochberg, Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *J. R. Stat. Soc. Ser. B Methodol.* **1995**, 57 (1), 289–300.
- (60) Calinski, T.; Harabasz, J. A Dendrite Method for Cluster Analysis. *Commun. Stat. - Theory Methods* **1974**, 3 (1), 1–27.
- (61) Davies, D. L.; Bouldin, D. W. A Cluster Separation Measure. *IEEE Trans. Pattern Anal. Mach. Intell.* **1979**, PAMI-1 (2), 224–227.
- (62) Rousseeuw, P. J. Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. *J. Comput. Appl. Math.* **1987**, 20, 53–65.
- (63) Moran, P. A. P. Notes on Continuous Stochastic Phenomena. *Biometrika* **1950**, 37 (1–2), 17–23.
- (64) Geary, R. C. The Contiguity Ratio and Statistical Mapping. *Inc. Stat.* **1954**, 5 (3), 115–141.
- (65) Belkin, M.; Niyogi, P. Laplacian Eigenmaps for Dimensionality Reduction and Data Representation. *Neural Comput.* **2003**, 15 (6), 1373–1396.
- (66) Shuman, D. I.; Narang, S. K.; Frossard, P.; Ortega, A.; Vandergheynst, P. The Emerging Field of Signal Processing on Graphs: Extending High-Dimensional Data Analysis to Networks and Other Irregular Domains. *IEEE Signal Process. Mag.* **2013**, 30 (3), 83–98.